



Improved Particle Swarm Optimization based Bidirectional-Long Short-Term Memory for Intrusion Detection System in Internet of Vehicle

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Abstract

An intelligent vehicle communication network referred to the internet of vehicles (IoV) has been developed by integrating the vehicle network (VN), the internet of things (IoT) and the vehicle-to-everything (V2X) technologies. However, malicious assaults may easily compromise the vehicle communication network. Protecting the integrity of data exchanged between vehicles in an IoV network demands an intrusion detection system (IDS). It is crucial to understand that the distribution of network traffic data is not uniform, and the presence of various forms of network attacks significantly impacts the functioning of IDS. Data imbalance is a prevalent issue in network traffic data and significantly affects the accuracy of detection. This research addresses this challenge by using the Synthetic Minority Oversampling Technique (SMOTE) to introduce a well-proportioned distribution of classes. However, the wide range of network threats is another important issue that contributes to the inadequate performance of intrusion detection models. The majority of existing Deep Learning (DL)-based IDS primarily focus on binary classification and do not achieve substantial detection of advanced attacks. On the other hand, DL models such as Long Short-Term Memory (LSTM), Convolutional Neural Network (CNN), Bidirectional-LSTM (B_LSTM), etc. encounter several challenges when it comes to constructing a particular framework by carefully choosing the suitable and optimum values of hyper-parameters. This research implements the Chaotic Particle Swarm Optimization (CPSO) technique to effectively optimize several important hyper-parameters of the B_LSTM model. These hyper-parameters include the learning rate, number of neurons in hidden layers, dropout rate, optimizer and activation functions of the B_LSTM layers. The objective is to improve the model attack detection accuracy performance. Implementation of the proposed B_LSTM + CPSO model using the “Car Hacking: Attack & Defence Challenge-2020” datasets demonstrate outstanding performance in detecting attacks in the IoV environment. The experimental findings have been compared with two other optimization strategies, namely the PSO and Firefly Algorithm (FA). Subsequently the findings are also compared to the performance of standard LSTM, LSTM + PSO, LSTM + FA, LSTM + CPSO and B_LSTM algorithms. The evaluation of the statistical comparisons demonstrates that the proposed B_LSTM + CPSO technique outperformed other investigated approaches in attack detection with an accuracy of about 99.94%. The model optimizes hyper-parameters including a learning rate of 0.001, number of hidden units of 128, dropout rate of 0.1, optimizer as ‘Adam’ and layer activation function as ‘relu’. This research presents a helpful resource for assuring the cyber security of the IoV.

Keywords IoV · IDS · Security · B_LSTM · LSTM · PSO · FA · CPSO · Optimization · Hyper-parameter

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1 Introduction

In recent years, technological evolution has greatly improved the transmission and communication features of the many devices we use in our daily routines. A wide variety of various gadgets now efficiently extend the reach of the internet to any region of the world. These devices offer a range of responsibilities via their ability to link with other devices. Components such as smartphones, automobiles, smart speakers, smart medical equipment and smart dispensing robots have the capability to connect to the internet and share data. The IoT facilitates a diverse range of services to consumers throughout the whole network architecture. The IoT is expected to increase levels of automation. The integration of the IoT with smart health, smart cities, smart agriculture [1], smart industry and smart transportation improves the efficiency and efficacy of these technologies. The primary goal of this intelligent environment is to enhance cost-effectiveness, efficiency, safety and resource utilization [2]. The transportation sector has successfully utilized the IoT to their advantage by creating novel approaches of lightweight and internet-based communication. The gradual implementation of wireless networks on roads and vehicles has allowed for the transmission of sensitive information between vehicles and infrastructure. To enhance the intelligence and safety of these vehicles they have been fitted with both outdoors and internal sensor systems. Several sensors like parking cameras and motion detectors are installed outside the car, while other sensors for fuel, brake and tire pressure are installed inside. The IoV refers to a specific category of the IoT that consists of automotive sensors that can exchange data over a wireless and mobile connection.

IoV integrates vehicle ad hoc networks (VANETs) with the IoT to enhance autonomous transport. [3]. The IoT incorporates the IoV to bring about a transformative change in the transportation sector. IoV is a network system that has emerged from the IoT and offers significant controllability, manageability and authenticity. Multiple vehicles, users, objects and networks enable many IoV cloud server systems to exchange information with their environment via diverse communication routes. IoV is widely recognized as an important IoT application in the field of automobiles, as shown by several studies [4]. Smart motor vehicle uses V2X connections to optimize road safety and enhance traffic congestion mitigation, therefore supporting advanced transportation and road mobility [5]. Vehicles communicate with each other, nearby infrastructure, personal devices, embedded sensors and the cloud through V2X communications. This includes vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), vehicle-to-pedestrian (V2P), vehicle-to-sensor (V2S) and vehicle-to-cloud (V2C) interactions with one another [6]. These interactions occur wirelessly using various wireless technology platforms, as shown in Fig. 1.

Commonly, V2X communications are used in two types of applications: (1) for safety including avoiding crashes or notifying drivers of potential collisions and (2) for non-safety applications such as navigation, theft prevention, or diverting attention [7]. When these interactions integrate with the capabilities of vehicles with on-board sensors to produce alerts and messages about obstacles, hazards, collisions and congestion, road safety and traffic management seen to improve significantly. Safe and dependable data exchange is essential for vehicle applications due to the vital nature of these systems, especially safety-critical systems. Still, vehicle networks are quite vulnerable to both external attacks in which the attacker is not a verified user and also insider attacks in which the attacker is a verified user. These vehicles might cause accidents and expose people to risk because of their malicious purpose, which can lead to excessive delays in transportation, forging and/or modifying safety-related information and other such related issues [8]. One significant issue is that the IoV is experiencing essential security issues because of the rapid increase in vehicles connected to the network and the more sophisticated network environment, which has led to a rise in adverse network attacks. There are serious risks to both its communication security and data security. A vehicle's security and dependability are at risk if it has been compromised by hacking, which might result in mortality or traffic accidents. As an example, hackers may remotely manipulate several automotive components such as the engine, brakes, accelerator, steering, etc. potentially disrupting or even damaging the vehicle's normal operation. Furthermore, by obtaining or altering vehicle data, hackers have authority to violate the privacy or interests of the individual who controls the vehicle.

The IDS can detect symptoms of network attacks in the IoV gathering network traffic information inside the IoV and then processing and analyzing this data to achieve a categorization of attack and non-attack samples. By implementing the IDS on cars or Roadside Units (RSUs), it becomes possible to analyze and evaluate the incoming communication data to the vehicle or RSU. If any anomalous or dangerous behavioral data is detected, the system has the capability to reject the data and prevent its transmission, thereby identifying assaults inside the IoV. IoV intrusion detection approach may be extensively categorized using both conventional Machine Learning (ML)- and DL-based techniques. Support Vector Machine (SVM) [9], Decision Tree (DT) and other classical ML techniques like Adaptive Boosting (AdaBoost), Random Forest (RF). manually extract features and classify attack and normal sample data. Subsequently addressing the massive and complicated intrusion detection information available on the IoV, these methods exhibit various limitations, including extended processing times and inadequate detection accuracy. Presently DL is the primary approach adopted by researchers for intrusion detection in the IoV. It is

Fig. 1 Communication overview of IoV environment

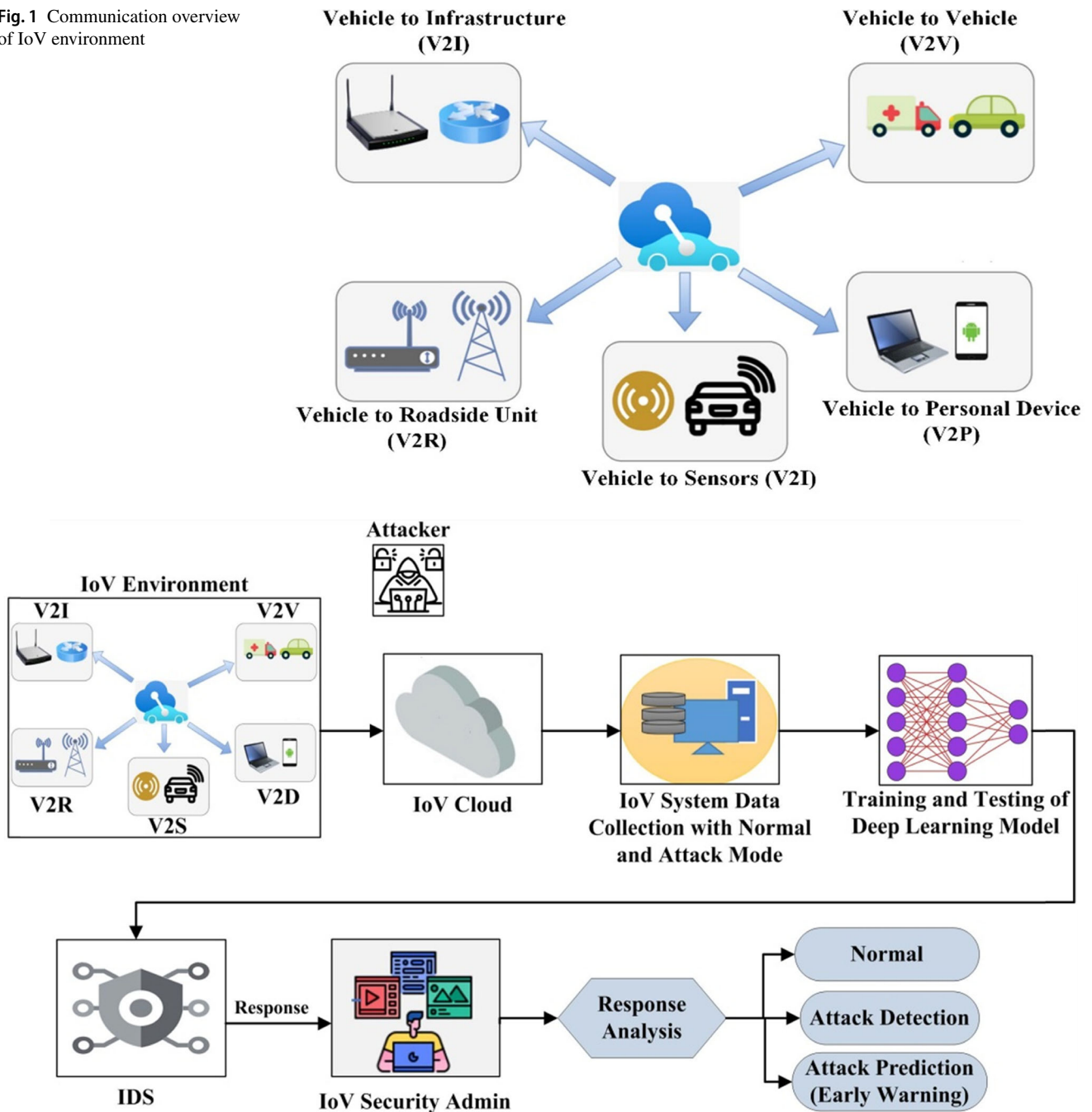


Fig. 2 General architecture of IoV IDS

possible to extract underlying characteristics from the intrusion sample data with the implementation of DL [10–13], which enables the detection of network intrusion reports. Several DL models including Generative Adversarial Network (GAN), Deep Belief Network (DBN), CNN, Recurrent Neural Network (RNN), LSTM and B_LSTM have been employed for developing intrusion detection approaches in the field of IoV. Figure 2 illustrates the IoV IDS framework.

1.1 Background Study

An effective strategy includes employing B_LSTM to detect several threats inside the IoV network. In order to effectively identify intrusions in the IoV environment, which involves interconnected vehicles communicating with each other and with infrastructure, it is essential to gather contextual dependencies from previous and upcoming inputs. Also, the declining gradient issue has a partial solution offered by

bidirectional processing. In traditional LSTM models, the back propagation over time approach may result in the vanishing gradients issue, which degrades the capacity to store connections over extended periods of time. The bidirectional technique integrates data in the opposite way to overcome this problem. Analyzing the model's performance using different hyper-parameters might improve the behavior of the DL algorithm. By fine-tuning hyper-parameters, models become more robust against variations in the training environment or input data, hence enhancing their performance consistently throughout the training process. Optimizing the hyper-parameters of the model may lead to improved accuracy, faster convergence and greater generalizability. IoV environments may undergo changes over time due to several reasons such as traffic patterns, network conditions and vehicle activities. By optimizing the model's hyper-parameters, the IDS can adapt to varying conditions, ensuring effective performance in all scenarios while reducing resource consumption and enhancing the reliability and user convenience of the IoV system.

Swarm-based algorithms are optimization methodologies that are inspired by the cooperative behavior seen in swarms of insects, birds and bees. Algorithms in this category include Ant Colony Optimization (ACO), PSO, FA and Bat Algorithm (BA). Hyper-parameter optimization in DL frequently utilizes these approaches to efficiently explore the parameter space with many dimensions. PSO and FA have shown impressive performance in intrusion detection for IoV-based applications by fine-tuning the DL models. However, PSO may experience premature convergence, a situation in which the particles converge on a poor solution before to fully traversing the complete search space. The performance of PSO may be influenced by its parameters, including the inertia weight, acceleration coefficients and swarm size. Optimizing these parameters to get the best possible performance in various problem domains and search spaces may be a difficult and time-consuming task. Also, slow convergence rates are a potential issue with the FA, particularly in optimization landscapes that are complicated and have multiple local optima. When compared to other methods, FA may take longer to converge which means it will take longer to optimize problems with many dimensions. In general, CPSO has several benefits compared to conventional PSO. Introducing chaotic sequences may enhance the efficiency of particle exploration in the search space by introducing randomness and unpredictability. This allows the algorithm to avoid getting trapped in local optimal solutions and instead explore a wider range of areas in the search space. CPSO exhibits more adaptability to parameter configurations as compared to conventional PSO. Traditional PSO sometimes requires careful adjustment of parameters, such as inertia weight and acceleration coefficients, to get optimum performance. In contrast, CPSO's chaotic dynamics inherently possess adaptability,

hence reducing the sensitivity to parameter configurations. CPSO is highly suitable for solving complicated and multi-dimensional optimization issues in IoV environment.

1.2 Research Contributions

The B_LSTM approach requires tuning of several hyper-parameters to achieve optimum performance. These include the number of hidden neuron units, the number of B_LSTM layers, learning rate, dropout rate and batch size. Improper selection of these hyper-parameters might result in slow-moving convergence, training fluctuation, or inadequate performance. Addressing this, a swarm-inspired (SI) CPSO approach is implemented to determine the B_LSTM hyper-parameters. This research compares the implementations of two fundamental SI optimization techniques PSO and FA. The main disadvantage of these optimizations is that PSO and FA may demonstrate slow speed convergence rates, especially in optimization problems that are high-dimensional or complicated and the algorithm's ability to explore and use the space effectively decreases. These disadvantages lead to slower convergence and often fewer possible solutions. To surpass the constraints of traditional swarm-based optimization, this research introduces CPSO which promotes exploration of various locations in the search space to achieve comprehensive coverage of the solution space. The incorporation of chaotic alterations adds randomness and unpredictability to the motions of particles, which prevents premature convergence and encourages the exploration of a wide range of unique and non-repetitive solutions. The presence of diversity may result in a more extensive investigation into the range of possible solutions as well as enhance overall performance. Following are the key highlights of this research:

- A modified swarm-based approach is proposed to improve the fine-tuning of hyper-parameters of the IDS in the IoV networks, which are distinctive by their dynamic and diverse nature.
- The extensively utilized B_LSTM framework has been implemented for detecting and understanding anomalies that communicate over time in the complex and dynamic data of the IoV network.
- A novel CPSO algorithm is employed to improve the hyper-parameters selection of B_LSTM, therefore improving detection accuracy and flexibility. This technique has great potential in boosting the security of IoV environment.
- Several ML approaches including DT, RF, logistic regression (LR) and AdaBoost are evaluated. In addition, significant experiments are performed with LSTM, B_LSTM networks, B_LSTM with PSO, FA and CPSO. The proposed B_LSTM model with CPSO algorithm produced

results showing considerable effectiveness of IDS in the IoV environment.

1.3 Current Research Scope

This present work is intended to extend the primary goal of improving attack detection in the IoV environment by implementing B_LSTM + CPSO approach. The research can offer several opportunities for further investigation and wider implementations. Some of the possibilities can be enlisted as follows.

- *Deployment in Multiple IoT Environments:* The proposed methodology presented in this research may be adapted to ensure the security of diverse IoT applications such as smart cities, industrial IoT and healthcare systems. The chaotic-based PSO approach's generalizability offers promising prospects for expanding the concept to other sectors where security is of highest priority.
- *Deployment in Real-World IoV:* Further evaluation of the proposed model will enable its implementation in real vehicle networks, therefore making valuable contributions to industrial applications in the automotive and cybersecurity sectors. The present research has the potential to be extended in order to evaluate the scalability and effectiveness of the model within practical applications therefore facilitating its real-world application.
- *Impact on Cross-Disciplinary Platform:* DL models are integrated into autonomous, financial and healthcare technologies. As a result, the proposed B_LSTM + CPSO technique may have an effect on these industries. The optimization of Bi-LSTM for attack detection by the chaotic-based PSO may result in its adoption in other industries requiring complex prediction and decision-making models.

The remaining sections of this article follow the proposed structure: Section 2 presents a comprehensive summary of the latest research in the specific area focused on improving security inside the IoV ecosystem. Section 3 describes an in-depth overview of the materials and methods for the present research. This section presents proposed anomaly-based IDS, along with its issue formulation. Section 4 of our research focuses on providing a comprehensive summary of the simulation setup that we used for evaluation of our model. Section 5 relates to the evaluation of our experimental analysis and the outcomes achieved, while sect. 6 serves as the conclusion section of the research.

2 Existing Studies

IDSs play a vital role in the continuously developing IoV ecosystem. They serve as cyber defenders, safeguarding interconnected vehicle networks against a wide range of intrusions. This section introduces a concise overview of the latest and advanced DL-based IDS methodologies specifically designed for the IoV environment.

To solve the issue of exchanging the sensitive CAN information, Aliyu et al. [14] built an in-vehicle network security system that was implemented by software-defined networking (SDN) and was designed on the blockchain. Within the detection model system, a RF model was created using a Federated Learning (FL) approach. This method aggregated partly trained models that were trained independently throughout the process to keep their data hidden. In order to enhance the ability of detecting attacks across multiple classes, they employed the Fourier transform to break down the cycle of CAN IDs from the data obtained from the CAN bus, in the frequency domain. They utilized the CAN-intrusion dataset (OTIDS) in their research. A detection accuracy of 98.10% was achieved by their proposed approach. Shams et al. [15] introduced a new DL model, Self-Attention Layer within a Convolutional Neural Network (SACNN), to recognize auditory data in large datasets of emergency vehicle sirens and road noise. The dataset included 200 3-second WAV-format audio files grouped into Emergency Vehicles (Ambulance and Firetruck sirens) and Traffic (ordinary traffic noises), each with 200 spectrogram images per file. Experimental findings showed SACNN outperformed other measures like accuracy, precision, recall and F1 score, while also improving computing economy. SACNN outperformed EfficientNet and 1D_CNN on a large dataset, with an average accuracy, precision, recall and F1 score of 1.00. Additionally, SACNN showed the maximum computational efficiency, averaging 0.20883 s per sample. SACNN effectively detected and classified emergency vehicle sirens and road noise in big datasets, impacting traffic management, public safety and noise pollution monitoring. SACNN showed promise to create an efficient real-time acoustic data detection system.

A ConvLSTM-based IDS for in-vehicle networks (IVNs) was designed to take utilization of the recurrence of the network message ID by Yang et al. [16]. The framework employed a proximal policy optimization (PPO)-based federated client selection (FCS) technique to reduce system complexity and maximize model accuracy during the training of federated ConvLSTM models. The modeling approach utilized IVN datasets and real-life IoV scenario situations. Implementing ConvLSTM significantly reduced model size and convergence time while keeping detection accuracy at 95% or higher, according to the results obtained by the authors. Alladi et al. [17] designed an IDS particularly aimed



to identify cyber-attacks in the IoV infrastructure. The system utilized DNN model running on a Mobile Edge Computing (MEC) server, which was linked to the RSU. The proposed system utilized two classification approaches, one of which built temporal patterns from network broadcast data. The second approach relied on categorizing network traffic by using visual representations of the time-based patterns. The authors demonstrated the efficacy of the suggested IDS by using a Raspberry Pi 3B. To achieve this, they utilized the Veremi Extension dataset to simulate four separate DL algorithms: MLP, CNN, LSTM and CNN-LSTM. This dataset consisted of traces collected from simulated VN samples. The CNN-LSTM model obtained a sequence classification accuracy of 99.42%.

A DL-based intrusion detection method was developed by Ashraf et al. [18] to identify possibly malicious network traffic in V2X interactions. To successfully detect both known and unknown assaults, their hybrid IDS implemented the LSTM autoencoder approach instead of signature-based detection. The primary network connections of vehicles were the target of this technology's intrusion detection skills. During testing the IDS's performance, the authors examined two datasets: one for intra-vehicle networks (IVN) and another for exterior vehicular networks (UNSW-NB15). On the IVN dataset, the suggested method correctly identified all types of assaults at an accuracy score of over 99%, while on the UNSW-NB15 dataset, it attained accuracy score of 98%. Specifically, for the IoV, Yang et al. [19] designed a multi-tiered hybrid IDS. The primary objective was to correctly detect and classify both known and novel assaults on internal and external vehicle networks. This IDS employed a combined approach of signature-based and anomaly-based methodologies. The authors evaluated their suggested IDS using two different datasets to determine its efficiency and effectiveness. Data related to car hacking, particularly the in-vehicle network was included in the initial set of records. The CICIDS2017 dataset included details on traffic from outside networks. The findings showed that the framework obtained an identification accuracy of 99.99% for various types of known attacks on the CAN-intrusion-dataset, which represented the IVN data. In addition, their suggested model attained a precision of 99.88% on the CICIDS2017 dataset, which specifically included data from external vehicular networks.

Khan et al. [20] proposed a hybrid IDS that accurately identified both inside and outside assaults in the IoV. The system employed a bloom filter and B_LSTM architecture both of which were derived from DNN to detect zero-day assaults. The effectiveness of the presented IDS was assessed using a dataset including samples of automobile hacking on a CAN and the UNSWNB-15 external vehicular network dataset. The results demonstrated that the suggested framework attained exceptional performance, surpassing most

cutting-edge methods with an accuracy rate of 98.88% for the UNSWNB-15 dataset and 99.11% for the automotive hacking dataset. Furthermore, the suggested system had the capability to detect previously unknown (zero-day) incidents occurring in IoV networks. The misbehavior detection system (MDS) designed by Hsu et al. [21] utilized DL techniques such as CNN and LSTM models to reconstruct a vehicle's location data from receiving Basic Safety Messages (BSMs). An SVM classifier was used to determine if the receiver vehicle had been hacked or not, based on the content of the communication that was received. The MDS used by authors participated in training and evaluation using the widely-accepted VEREMI dataset, which encompasses several forms of fraudulent activity in a vehicle environment. The research results demonstrated that the proposed algorithm exhibited a high level of precision in identifying 95.37% of the specified cars.

Halbouni et al. [22] proposed a hybrid model for IDS called CNN-LSTM. This model combined CNNs with LSTM in their study. The CNN-LSTM methodology was employed on WSN-DS, UNSW-NB15 and CIC-IDS-2017 datasets, and the obtained accuracy scores were 99.67%, 94.53% and 99.64% respectively. Javeed et al. [23] designed a hybrid approach by using Cuda-deep neural network gated recurrent unit (CuDNNGRU) and Cuda-deep neural network LSTM (CuDNNLSTM) algorithms to create a model with the ability to identify distributed denial of service (DDoS) assaults. According to their research, the CuDNNLSTM + CuDNNGRU model attained a 99.74% accuracy rate in only 9.33 milliseconds. However, they have not disclosed the details about their methodology for extremely managing imbalance data. Table 1 describes the summary of existing studies of IoV security.

Several research studies have presented multiple strategies concerning the design of novel IDS techniques for identifying unauthorized access in IoV networks. However, existing methodologies considered in the literature include certain constraints. Initially, most suggested methods only focused on identifying a restricted range of cyber-attacks. Several datasets utilized in current research inadequately represent the real-life scenario of the IoV. Many present approaches lack sufficient recommendations for handling severely unbalanced data, a frequent occurrence in intrusion detection datasets. Existing studies utilized outdated datasets such as CIC-IDS-2017, UNSW-NB15 and WSN-DS, which consisted of traditional, fixed, unvalidated and non-replicable instances of malicious behaviors. Multiple studies have not offered significant focus on processing time, CPU time and system resources. However, the IoV network plays a vital role since it is responsible for performing time-critical tasks while both identifying and reducing any risks to the IoV system.



Table 1 Summary of existing studies

Ref	Dataset utilized	Model	Performance	Benefits of the study	Limitation of the study
[14]	CAN-BUS	RF-based federated learning	Accuracy:98.10%	This study minimized the amount of communication required in the IoV network This research offered a distributed model training approach	Accuracy of IDS can be improved designed for IoV security
[15]	Auditory dataset	Self-attention CNN	F1 score:100.00%	This study focused on specific type of noise identification	Proposed model should be evaluated on another dataset Fine-tuning the model could improve the generalization of the study on another related domain
[16]	IVN	ConvLSTM	Accuracy:95.00%	This study experimented with the real-time intrusions in IoV environment	Model performance is very poor. Also, the model should experiment on larger dataset
[17]	VeReMi	CNN-LSTM	Accuracy:99.42%	This study mainly focused on classification of real-time attack in RSU of IoV environment	Proposed model suffers high-cost computation complexity
[18]	UNSW-NB15	LSTM-based Autoencoder	Accuracy:99.00%	Aim of this study was to identify the inter-vehicle networks attacks of IoV	Model should implement on diversity of IoV network attacks To build generic IDS, model should train on more new attacks from both internal and external vehicle network
[19]	CICIDS2017	Multi-tiered hybrid IDS	Accuracy:99.88%	This study offers real-time attack detection from both internal and external IoV network	The proposed model having high computational cost and the study conducted on older dataset
[20]	UNSWNB-15	B_LSTM	Accuracy:98.88%	This proposed DL model was designed for only external IoV network attack	This study does not consider the internal IoV network attack. Also, the achieving accuracy is not satisfactory
[21]	VEREMI	SVM	Accuracy:95.37%	This study was conducted for a specific vehicle attack	Accuracy and other performance are not satisfactory Advanced model can be implemented for better attack detection rate
[22]	CIC-IDS-2017	CNN-LSTM	Accuracy:99.64%	Aim of this study was to offer real-time IoV attack detection	High computational cost and other performance like F1 score is not satisfactory as compared to accuracy score for attack detection
[23]	CICDDoS 2019	CuDNNLSTM + CuDNNGRU	Accuracy:99.74%	The experimental results showed that the proposed model had a high detection rate of over 99% for various types of attacks including DDoS	The experiment was not conducted on a new attack dataset for IoV



Table 2 Details of 2020 car hacking dataset

“2020 Car hacking” dataset	Total samples	Attack records	Normal records
Final_stationary	1,270,310	179,998	1,090,312
Preliminary_driving	3,672,151	299,408	3,372,743
Preliminary_stationary	3,752,046	393,836	3,358,210

Table 3 Detailed feature information of 2020 car hacking dataset

Features	Feature meaning	Feature value type
TS	CAN Bus employ timestamps to allocate a certain time. A communication is shown along with its timestamp, which indicates the exact time that it was either transmitted or received	Continuous
ID1	The identification field serves the purpose of identifying the respective CAN frame	Continuous
DLC	The data length code in control field of 11-bits mode of CAN message serves to indicate bytes value of the data field within the range of 0 to 8. A DLC value of 0 describes a specific kind of message frame known as Remote Frame. Remote frames serve the purpose of requesting data from a designated node inside the network, without transmitting any more information	Continuous
Data	The Data field refers to the payload data that contains the actual data	Continuous
Subclass	Replay and fuzzing, spoofing, flooding and replay	Categorical
Class	Attack or normal	Categorical

The objective of this research is to address these drawbacks by proposing a novel and efficient DL system. The framework integrates properly chosen hyper-parameters to reduce the execution time for precisely detecting various malicious IoV attacks. The integration of chaotic behaviors into the PSO algorithm has the advantage of avoiding local optima by introducing randomization into the search procedure. Effective adaptation to local optima is crucial in IoV security settings, where the optimization environment might be complex and unpredictable. This flexibility is essential for determining hyper-parameter setups that present a dependable and secure B_LSTM model. The CPSO algorithm enhances the generalizability of B_LSTM models to unfamiliar IoV security environments by effectively exploring the hyper-parameter space and avoiding premature convergence to incompatible approaches.

3 Materials and Methods

This section describes dataset preparation details, overview of the B_LSTM Architecture framework, PSO and CPSO optimization algorithms.

3.1 Materials

This sub-section presents a detailed overview of the dataset including its characteristics, features and pre-processing adopted for the simulation.

3.1.1 Dataset

In this research we have implemented the datasets collected during the 2020 Car Hacking [24]: The final stationary dataset, the preliminary driving dataset and the preliminary stationary dataset are the three separate datasets that are used. For each dataset, the features are common. All the independent features have continuous values, whereas the target attribute has categorical values. Table 2 describes a detailed overview of all three datasets including the total number of samples for both attack and normal types. The dataset consists of features such as TS (Timestamps), ID1 (Arbitration-ID), DLC (Data Length Code), Data (CAN Data Field) and Class (Normal or Attack). The features common to each dataset along with their corresponding meanings are summarized in Table 3.

3.1.2 Data Pre-Processing

- *Missing Value Imputation:* The first step of proposed framework involves the process of missing value imputation. Imputation is a technique used to replace missing or null values in data. The imputation approach used in the present research is backward filling. The subsequent valid observation are used to fill the gap.
- *SMOTE Oversampling:* Data imbalance refers to an environment when the class distributions in a dataset are unequal. Imbalanced data leads to bias as the methodology leads to prioritize the dominating classes, resulting

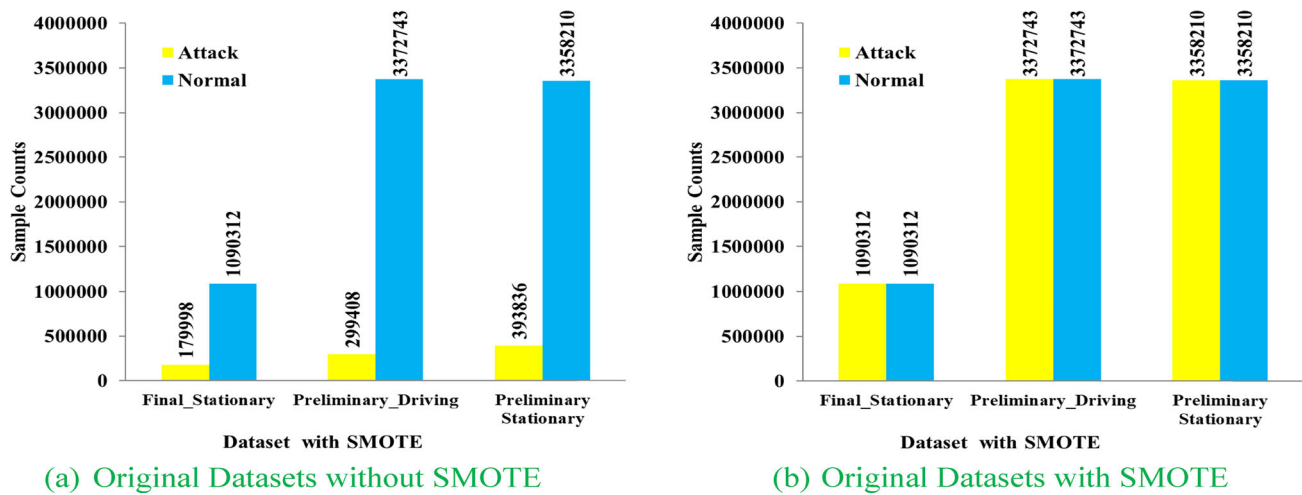


Fig. 3 Graphical distributions of datasets

Table 4 Details of training and testing sizes for 2020 car hacking dataset

“2020 Car Hacking” Dataset	Total Samples	Training Samples (80%)	Testing Samples (20%)
Final_stationary	2,180,624	1,744,500	436,124
Preliminary_driving	6,745,486	5,396,389	1,349,097
Preliminary stationary	6,716,420	5,373,136	1,343,284

in a greater possibility of failing to differentiate minority class samples. A well-balanced dataset enhances the effectiveness of the model. The under-sampling approach (SMOTE) is employed with the B_LSTM + CPSO model to overcome the challenges of imbalanced samples. Figure 3a, b depict the graphical representation of a dataset, comparing its outcomes with and without the implementation of the SMOTE technique.

- **Normalization:** Normalization is the subsequent phase in the pre-processing stage. Data normalization aims to eliminate anomalies or outliers from the data while also reducing variations in the data. Time-series data must undergo a normalizing method to decrease the model’s dependence on large fluctuations in data. The MinMax Scaler is used to standardize the data in the current research.
- **Data Splitting:** The final phase of pre-processing involves dividing the dataset into training, validation and testing data. A DL approach is examined using the training dataset. During each iteration, the validation dataset is used to evaluate the adjustment of hyper-parameters. The testing dataset is implemented to evaluate and compare the findings of the hybrid models CPSO with B_LSTM, PSO with B_LSTM and FA with B_LSTM to choose the best approach. The purpose of dividing the dataset is to prevent model overfitting and enhance the reliability and robustness of the DL approach. For this research, 80% of the data samples are allocated for training purposes, while the remaining 20% are used for validation. Table 4 illustrates

the total number of samples for each dataset, as well as their training size and testing size for evaluating the methodologies.

3.2 Methods

This sub-section describes an overview of the B_LSTM Architecture framework, PSO and CPSO optimization algorithms.

3.2.1 PSO

The PSO algorithm was first presented by scientists Eberhart and Kennedy in 1995 [25]. This algorithm is derived on the social behaviors of birds and is designed to mimic evolutionary processes. The PSO algorithm improves problem-solving skills by modeling them after the coordinated actions of biological groups. The benefits include simple explanation, straightforward execution, minimal parameters customization, rapid convergence and cost-effective computational expense. There are no demanding memories or CPU speed requirements. In particle optimization, each potential solution to the issue is treated as a “particle” in a space of N dimensions. Within solution space, the particle travels at a certain velocity and in a specific direction. Every particle in the system makes use of two global variables: one that indicates the best possible location for the particle specifically (local particle best) and another that indicates the greatest



possible location for all particles together (global particle best). In an N -dimensional search space, the particle population $\theta = (\theta_1, \theta_2, \dots, \theta_N)$ consists of K particles. The i th particle's position is represented as $\theta_i = (\theta_{(i,1)}, \theta_{(i,2)}, \dots, \theta_{(i,N)})$ and its velocity is indicated as $\vartheta_i = (\vartheta_{(i,1)}, \vartheta_{(i,2)}, \dots, \vartheta_{(i,N)})$. The individual best value is represented by $P_i^{LBest} = (P_{(i,1)}^{LBest}, P_{(i,2)}^{LBest}, \dots, P_{(i,N)}^{LBest})$, the global best value of the particle population is represented by $P_i^{GBest} = (P_{(i,1)}^{GBest}, P_{(i,2)}^{GBest}, \dots, P_{(i,N)}^{GBest})$. During iteration ' $t + 1$ ', the particle adjusts its velocity and location using Eqs. (1) and (2).

$$\vartheta_{(i,D)}^{t+1} = W_{pso} * \vartheta_{(i,D)}^t + C1 * R1(P_{(i,D)}^{LBest(t)} - \theta_{(i,D)}^t) + C2 * R2(P_{(i,D)}^{GBest(t)} - \theta_{(i,D)}^t) \quad (1)$$

$$W = \begin{cases} W_{pso(min)} + \frac{(W_{pso(max)} - W_{pso(min)})(F_{objective} - F_{objective(min)})}{(F_{objective(avg)} - F_{objective(min)})}, & F_{objective} \leq F_{objective(min)} \\ W_{pso(max)}, & F_{objective} > F_{objective(min)} \end{cases}$$

$$\theta_{(i,D)}^{t+1} = \theta_{(i,D)}^t + \vartheta_{(i,D)}^{t+1} \quad (2)$$

where $\vartheta_{(i,D)}^t$ = Velocity of particle and $\theta_{(i,D)}^t$ = Position of particle

Both $R1$ and $R2$ are random variables with a uniform distribution and values between 0 and 1. The maximum step size is determined by two positive constants called acceleration coefficients, $C1$ and $C2$. The inertia weight, W_{pso} , determines how much of an impact a particle's past velocity has on its present velocity. Using its previous velocity and the distance between its present position and the best historical locations of itself and its neighbors, the new velocity is computed using Eq. (1) in the conventional PSO. Typically, each component in ϑ_i may be restricted to the range $[-\vartheta_{max}, \vartheta_{imax}]$ to prevent particles from straying too far from the search area. The particle moves toward a new place as defined by Eq. (2). The procedure continues until a stopping requirement is met.

3.2.2 CPSO

3.2.2.1 Adaptive Weight Factor (AWF) Effective management of local exploitation and global exploration are essential for efficiently identifying the optimal solution in PSO [26]. The performance of PSO is significantly influenced by its parameters. The initial component of Eq. (2) signifies the influence of the velocity from the previous iteration, which transfers the necessary acceleration for particles to travel through the whole of the search region. How much the past

velocity affects the current velocity is controlled by the inertia weight W ? The value of W_{pso} determines the balance between exploitation and exploration in PSO. Therefore, precise management of the inertia weight is crucial for correctly and effectively determining the optimal solution. With a smaller inertia weight, the current search area may be fine-tuned, while a larger inertia weight promotes widespread exploration. Shi and Eberhart [27] improved the performance of the PSO by using a linearly variable inertia weight that changes from 0.9 to 0.4 during the generations. We need to adjust the parameter W_{pso} in real-time depending on the optimum values of particles to determine a balance between employing known information and exploring new possibilities. The new W_{C_pso} is calculated as new AWF represented in Eq. (3).

$$W_{C_pso} = W \quad (3)$$

where $W_{pso(max)}$ and $W_{pso(min)}$ denotes the upper and lower values of W_{pso} , $F_{objective}$ is the fitness value of the particle and $F_{objective(avg)}$ and $F_{objective(min)}$ are the average and minimum fitness values of all particles, respectively. Following Eq. (3), the value of W_{pso} is adjusted based on the particle's fitness value. As a result, particles with average or higher objective values will be disrupted, while those with lower values will be sheltered. Best particles focus on exploiting local search to improve findings, whereas poor particles tend to explore space with massive modifications. AWF offers an effective method to preserve population variety and ensure strong convergence potential.

3.2.2.2 Chaotic Search Strategy Chaotic mechanisms are included into the PSO with AWF to enhance search behavior and prevent getting stuck in local optima. In order to construct a hybrid PSO, He et al. used the logistic equation [28], which exhibits sensitive dependence on initial conditions. The logistic equation can be expressed as in equation (4).

$$X_{k+1} = \mu * X_k(1 - X_k), 0 \leq X_0 \leq 1 \quad (4)$$

where, μ denotes control parameter, X is a variable and k is a non-negative integer. The equation shows chaotic dynamics when $\mu = 4$ and $X_0 \neq [0, 0.25, 0.5, 0.75, 1]$. The system has a notable degree of sensitivity to its initial factors, which is a fundamental attribute of chaos. Even a little difference in the initial state of the chaotic variable might lead to a sig-

nificant divergence in its long-term evolution. The direction of the unstable variable may exhibit exponential behavior throughout the whole search space. These specific variables display properties such as ergodicity, pseudo-randomness and irregularity. An explanation of the chaos local search process could have been introduced by the following Eq. (5).

$$ch_X_i^{(t+1)} = 4 * ch_X_i^{(t)} (1 - ch_X_i^{(t)}), i = 1, 2, 3, \dots, k \quad (5)$$

where ch_X_i is the i^{th} chaotic parameter and t present the iteration number. Obviously, $ch_X_i^{(t)}$ is distributed in the range (0, 1.0) under the conditions that the initial

$ch_X_i^{(t)} \in (0, 1)$ and that $ch_X_i^{(t)} \neq [0.25, 0.50, 0.75]$. So, in this research, the proposed approach is to implement a two-phase iterative technique named CPSO which combines the previously described PSO with AWF and chaotic search strategy. The current implementation utilizes PSO with AWF for global exploration and chaotic search strategy for locally directed search (exploitation) of the solutions introduced by PSO. Algorithm 1 displays the CPSO approach. The velocity update Eq. (6) for CPSO may be constructed based on the descriptions described above.

$$\begin{aligned} \vartheta_{(i,D)}^{t+1} = & W_{C_pso} * \vartheta_{(i,D)}^t + C1 * ch_X \left(P_{(i,D)}^{LBest(t)} - \theta_{(i,D)}^t \right) \\ & + C2 * (1 - ch_X) \left(P_{(i,D)}^{GBest(t)} - \theta_{(i,D)}^t \right) \end{aligned} \quad (6)$$

Algorithm 1. CPSO

-
1. Declare the essential parameters of CPSO
'K': Total no. of Particles, 'MAX_ITR': Maximum iterations
'C1', 'C2': Acceleration coefficients, 'N': Dimensional search space
'F_{objective}': Fitness Function
 2. Randomly generate the particle position $\theta_{(i,d)}$, ($i = 1, 2, 3, \dots, K$), ($d = 1, 2, \dots, N$)
 3. Randomly generate the particle velocity $\vartheta_{(i,d)}$, ($i = 1, 2, 3, \dots, K$), ($d = 1, 2, \dots, N$)
 4. Compute the fitness value of each particle by $'F_{objective}'$ function
 5. Set all computed fitness value of each particle as individual best value $'P_i^{LBest}'$
 6. Set best computed fitness value among all population as global best value $'P_i^{GBest}'$
 7. while ($itr < MAX_ITR$) do
 8. set the new $'W_{C_pso}'$ by applying equation (3)
 9. for $i = 1$ to K , do
 10. update the velocity $\vartheta_{(i,d)}$ for each particle $'i'$ by applying equation (6)
 11. update the position $\theta_{(i,d)}$ for each particle $'i'$ by applying equation (2)
 12. compute fitness of new particle position $\theta_{(i,d)}$ by applying $'F_{objective}'$
 13. if current fitness value $> P_i^{LBest}$
 14. $P_i^{LBest} = \text{current fitness value}$
 15. End if
 16. if $P_i^{LBest} > P_i^{GBest}$
 17. $P_i^{GBest} = P_i^{LBest}$
 18. End if
 19. End for
 20. End while
 21. Obtain the optimal $'P_i^{GBest}'$
-



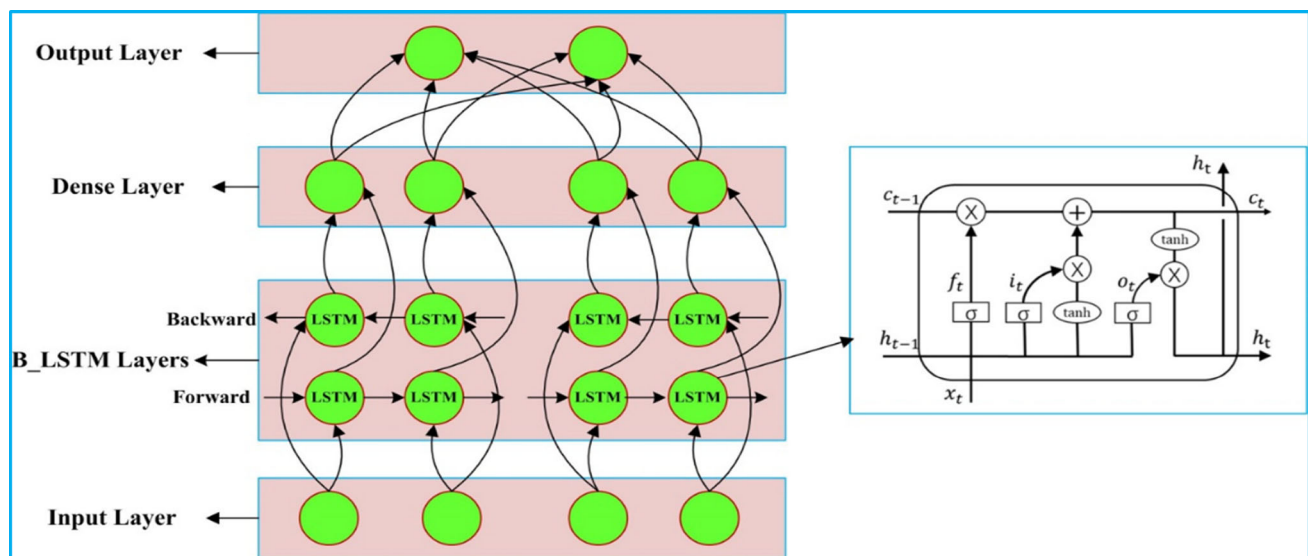


Fig. 4 B_LSTM architecture

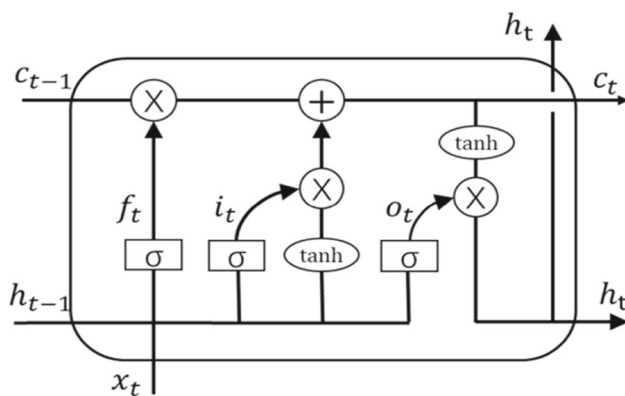


Fig. 5 LSTM cell structure

3.2.3 B_LSTM

This section explains the structure of the B_LSTM neural network, which has been developed to detect malicious attacks in the IoV network. The framework includes an output layer, a dense layer and several B_LSTM layers, illustrated in Fig. 4. The model's input includes a fixed-length vector which is a vector representation of the dataset characteristics based on investigation. The dense layer compresses the output vector dimension produced from the B_LSTM layer. The output layer receives the vector from the dense layer and produces the probability for classification. The B_LSTM layer is the crucial component of the model. The model is constructed by layering several LSTM cells. The LSTM cells are organized in a sequential fashion and analyze training samples in both the backward and forward directions. An LSTM creates and preserves a memory cell over its entire lifespan, which is a crucial component of the LSTM. Figure 5 demonstrates the

interaction between the prior cell state (C_{t-1}), the prior cell value (h_{t-1}) and the current input (X_t) in determining which components of the internal state vector should be updated, retained or disregarded. Three gates are implemented to control the changes in the state of the memory cell. The forget gate (f_t) selects which information to exclude from the cell state, the input gate (I_t) chooses which new information to keep in the cell state and the output gate (O_t) determines the specific output to generate. Following are Eqs. (7) to (9) of all three gates, including their current cell status (C_t) and current cell output (h_t) shown in Eqs. (10) and (11).

$$I_t = \sigma(W_{hi} * h_{t-1} + W_{Xi} * X_t + b_{input}) \quad (7)$$

$$f_t = \sigma(W_{hf} * h_{t-1} + W_{Xf} * X_t + b_{forget}) \quad (8)$$

$$C_t = \tanh(W_{hc} * h_{t-1} + W_{Xc} * X_t + b_{current}) \quad (9)$$

$$O_t = \sigma(W_{ho} * h_{t-1} + W_{Xo} * X_t + b_{output}) \quad (10)$$

$$h_t = O_t \tanh(C_t) \quad (11)$$

where ' b_{input} ', ' b_{forget} ', ' $b_{current}$ ' and ' b_{output} ' are bias vectors. ' W_{Xi} ', ' W_{hi} ', ' W_{Xf} ', ' W_{hf} ', ' W_{hc} ', ' W_{Xc} ', ' W_{Xo} ' and ' W_{ho} ' are the weight matrix of corresponding state.

One forward LSTM layer and one backward LSTM layer combine to form each B_LSTM layer, which results in the output vector ' h_t ' being divided into forward output ' $\vec{h}_t$ ' and backward output ' $\overleftarrow{h}_t$ '. The forward output ' $\vec{h}_t$ ' is computed iteratively from time 0 to time ' T ' using input in a positive

sequence, whereas the backward output $\overleftarrow{h}_t$ is calculated from time T to time 0 using the reversed input. Equations (12) and (13) describe the particulars of the equation, where the weight matrices are denoted as $W_{x\overrightarrow{h}}$, $W_{\overrightarrow{h}\overrightarrow{h}}$, $W_{x\overleftarrow{h}}$ and $W_{\overleftarrow{h}\overleftarrow{h}}$ while the bias vectors are represented by $b_{\overrightarrow{h}}$ and $b_{\overleftarrow{h}}$.

$$\overrightarrow{h}_t = \sigma \left(W_{x\overrightarrow{h}} X_t + W_{\overrightarrow{h}\overrightarrow{h}} \overrightarrow{h}_{t-1} + b_{\overrightarrow{h}} \right) \quad (12)$$

$$\overleftarrow{h}_t = \sigma \left(W_{x\overleftarrow{h}} X_t + W_{\overleftarrow{h}\overleftarrow{h}} \overleftarrow{h}_{t+1} + b_{\overleftarrow{h}} \right) \quad (13)$$

3.2.4 Proposed CPSO with B_LSTM

The proposed model is an integration of B_LSTM and CPSO (Algorithm 2). Unlike conventional LSTM that analyze input sequences in a unidirectional manner (from past to future), B_LSTM simultaneously process sequences in both directions. The initial configuration of the parameters of the B_LSTM neural network significantly influences the network's efficiency. In this research, the CPSO approach has been employed to enhance the crucial parameters of the B_LSTM network. Hyper-parameter tuning is the process of choosing the combination of hyper-parameters that result in the best performance for the B_LSTM model. The hyper-parameters consist of the dropout rate (θ_{dr}), learning rate (θ_{lr}), hidden units (θ_{hu}), layer activation function (θ_{af}) and optimizers (θ_{of}). Figure 6 depicts the flowchart of the CPSO_B_LSTM classification model. The proposed Fig. 7 illustrates the architecture of the proposed framework into five distinct processes: data collecting, data pre-processing, hyper-parameter initialization, modeling using CPSO_BLSTM and model evaluation.

3.2.5 Hyper-parameter Initialization

The purpose of the parameter initialization procedure is to assign an initial setting to the B_LSTM DL model. The CPSO, FA and PSO algorithms have been employed to optimize these hyper-parameters. The hyper-parameters that will be employed are:

- **Dropout rate (θ_{dr}):** Introducing a diverse dropout rate might enhance the performance of the B_LSTM approach by improving its ability to generalization.
- **Learning rate (θ_{lr}):** Learning rate is a hyper-parameter that adjusts the weights and biases of the model. Optimal model performance relies heavily on selecting an adequate learning rate, since it directly impacts the weighting and biases of the model.
- **Hidden units (θ_{hu}):** Determining the optimal number of neurons for each layer of the B_LSTM network is a

challenging problem. Insufficient neurons count in the B_LSTM affects its ability to retain the necessary information for producing optimum predictions. Conversely, if the number of neurons is very large, LSTM will exhibit overfitting on the training data.

- **Layer activation function (θ_{af}):** The selection of the activation function for the layers may have a substantial influence on the performance and behavior of the network. Activation functions are used for integrating nonlinearity into the network, which allows the network to learn and understand complicated patterns in the input. The selection of the activation function in a B_LSTM network can significantly impact its performance.
- **Optimizers (θ_{of}):** Inside a neural network, the optimizer has the responsibility of minimizing an objective function. By combining the optimizer and learning rate, the model's performance may be enhanced.

The hyper-parameter configuration space (θ) is a functional space with ' n ' dimensions that encompasses all potential combinations of hyper-parameters for the given B_LSTM network. A configuration vector is a vector that exists in the hyper-parameter configuration space and represents a distinct set of hyper-parameters for a B_LSTM network. The setting space for hyper-parameters may be vast, leading to a high demand on system resources. Therefore, the configuration space corresponds to the data presented in Table 5 of this research.

3.2.6 Modeling with CPSO_BLSTM

The implementation of CPSO algorithm starts with a swarm consisting of 16 particles and runs for a total of 50 iterations. The acceleration coefficients c_1 , c_2 have been assigned a value of 1.2. The search space bounds for each hyper-parameter are established (Table 5) and the particles (θ_i) are then randomly initialized in this specified search space, where $\theta_i = (\theta_{dr}, \theta_{lr}, \theta_{hu}, \theta_{af}, \theta_{of})$. Initially, the fitness objective function F_{b_lstm} is evaluated on θ_i . After computing the fitness of each particle by training the F_{b_lstm} with the relevant hyper-parameters and calculating the validation accuracy, the individual best position (P_i^{LBest}) and global best position (P_i^{GBest}) are initialized. Each iteration involves calculating the fitness of each particle $F_{b_lstm(i)}$ and comparing it to the best-fit values, both locally and globally. Particle fitness values are updated if they outperform their personal or global bests. Subsequently, the velocities ($v_{(i,d)}$) and locations ($\theta_{(i,d)}$) of the particles are modified according to the CPSO Eq. (6), using an inertia weight calculated from Eq. (3). The new location $\theta_{new(i,d)}$ of each particle is limited to the specified boundaries for each hyper-parameter. The CPSO method produces a collection of hyper-parameters that correspond to the global best position P_i^{GBest*} , achieving



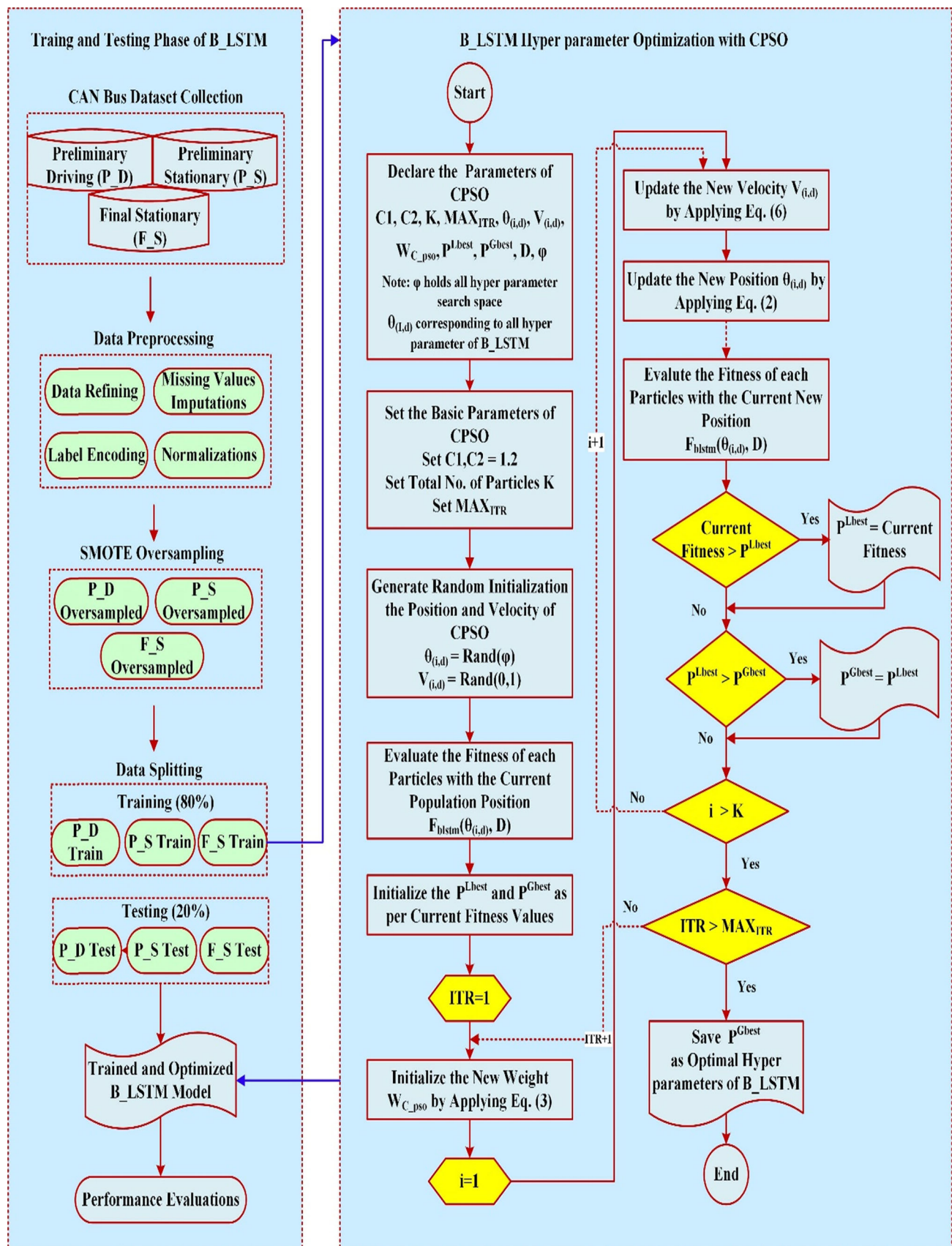


Fig. 6 Flowchart of proposed B_LSTM + CPSO algorithm



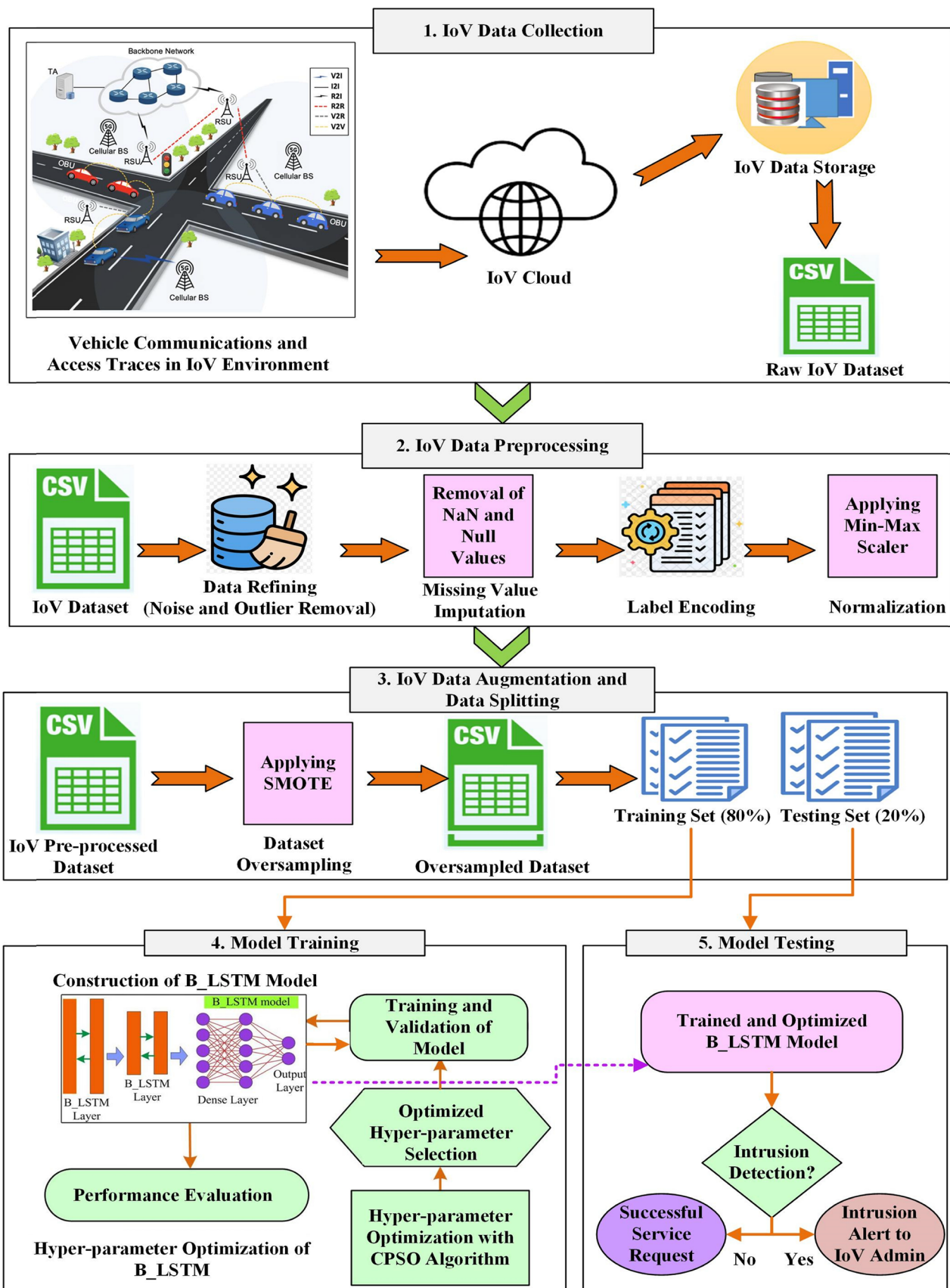


Fig. 7 Architecture of proposed framework for attack classification in IoV environment

the maximum validation accuracy after a specified number of iterations. The optimized hyper-parameters of the B_LSTM model using CPSO, as well as additional comparison strategies such as FA and PSO are presented in Table 6.

3.2.7 B_LSTM Model Evaluation

Using the optimized hyper-parameters (P_i^{GBest*}) that were discovered throughout the CPSO process, this research constructs the final CPSO using the B_LSTM model. The

training dataset is employed to train the model for 50 epochs with an optimal batch size of 160. A different validation dataset is utilized to confirm the trained model's performance once training is finished. Finally, optimized model has been tested on an independent dataset that was not impacted by any of the previous optimization or development stages.

Algorithm 2. CPSO with B_LSTM

1. Declaration:
 $\theta_{(i,d)}$ = Particle position, $\vartheta_{(i,d)}$ = Velocity of particles,
 F_{blstm} = Objective function, P_i^{LBest} = Local best of particle,
 P_i^{GBest} = Global best of particle, W_{c_ps0} = Weight value
 2. Input:
 MAX_{itr} = Maximum iterations, K = Total No. of particles,
 $C1, C2$ = Acceleration coefficients, D = Dataset,
 φ = Potential search space of hyper – parameters
 3. Output:
 P_i^{GBest} = Optimal best set of hyper – parameters
 4. Start
 5. for $i = 1$ to K , do
 6. Initializing particle position by generating random value within potential search space as $\theta_{(i,d)} = rand(\varphi)$
 7. Initializing particle velocity by generating random value as $\vartheta_{(i,d)} = rand()$
 8. Evaluate the objective function with current particle position
 9. as $f_i = F_{blstm}(\theta_{(i,d)}, D)$
 10. Set the local best of each particle as current fitness of particle $P_i^{LBest} = f_i$
 11. End for
 12. Find the max fitness value among all particles and set the best fitness as global best $P_i^{GBest} = \max(P_i^{LBest})$
 13. while ($itr < MAX_{itr}$) do
 14. Set the new W_{c_ps0} by applying equation (3)
 15. for $i = 1$ to K , do
 16. Update velocity $\vartheta_{(i,d)}$ for each particle 'i' by applying equation (6)
 17. Update position $\theta_{(i,d)}$ for each particle 'i' by applying equation (2)
 18. Evaluate fitness value of new particle position $\theta_{(i,d)}$
 19. by applying $f_i = F_{blstm}(\theta_{(i,d)}, D)$
 20. if $f_i > P_i^{LBest}$
 21. $P_i^{LBest} = f_i$
 22. End if
 23. if $P_i^{LBest} > P_i^{GBest}$
 24. $P_i^{GBest} = P_i^{LBest}$
 25. End if
 26. End for
 27. End while
 28. Save the optimal P_i^{GBest}
-



Table 5 Potential ranges of the hyper-parameters of B_LSTM model

B_LSTM Hyper-parameters	Potential search space
Dropout rate (θ_{dr})	[0,0.1,0.2,0.3,0.4,0.5]
Learning rate (θ_{lr})	[0.1,0.01,0.001,0.0001,0.02,0.03,0.04]
Hidden units (θ_{hu})	[8,16,32,64,128,256]
Layer activation function (θ_{af})	['relu', 'tanh', 'sigmoid']
Optimizers (θ_{of})	['Adam', 'Nadam', 'Adamax', 'Adagrad', 'Adadelta', 'RMSprop', 'SGD']

Table 6 Details of optimized hyper-parameters of B_LSTM on CAN bus datasets

Dataset Implemented	B_LSTM + PSO	B_LSTM + FA	B_LSTM + CPSO
Preliminary stationary	$[\theta_{lr} = 0.02, \theta_{dr} = 0.3, \theta_{hu} = 128, \theta_{af} = \text{'relu'}, \theta_{of} = \text{'Adam'}]$	$[\theta_{lr} = 0.04, \theta_{dr} = 0.2, \theta_{hu} = 128, \theta_{af} = \text{'relu'}, \theta_{of} = \text{'Nadam'}]$	$[\theta_{lr} = 0.001, \theta_{dr} = 0.2, \theta_{hu} = 64, \theta_{af} = \text{'relu'}, \theta_{of} = \text{'Adam'}]$
Preliminary Driving	$[\theta_{lr} = 0.04, \theta_{dr} = 0.1, \theta_{hu} = 256, \theta_{af} = \text{'tanh'}, \theta_{of} = \text{'Adam'}]$	$[\theta_{lr} = 0.001, \theta_{dr} = 0.4, \theta_{hu} = 64, \theta_{af} = \text{'relu'}, \theta_{of} = \text{'Adam'}]$	$[\theta_{lr} = 0.001, \theta_{dr} = 0.1, \theta_{hu} = 128, \theta_{af} = \text{'relu'}, \theta_{of} = \text{'Adam'}]$
Final stationary	$[\theta_{lr} = 0.01, \theta_{dr} = 0.3, \theta_{hu} = 64, \theta_{af} = \text{'relu'}, \theta_{of} = \text{'Nadam'}]$	$[\theta_{lr} = 0.01, \theta_{dr} = 0.3, 128, \theta_{af} = \text{'relu'}, \text{'Adam'}]$	$[\theta_{lr} = 0.001, \theta_{dr} = 0.1, \theta_{hu} = 128, \theta_{af} = \text{'relu'}, \theta_{of} = \text{'Adam'}]$

4 Experimental Configuration

The experiment has been performed using a Python notebook on GPU servers offered by Google Colab utilizing the TensorFlow and Keras frameworks. The computational configuration for this experiment included a 32 GB RAM module, a 3.20 GHz Intel Core i7 central processing unit (CPU), a 64-bit Windows 11 operating system and a GeForce GTX 1050 graphics processing unit (GPU) by NVIDIA. There are several packages available in the PYTHON programming language such as, Numpy, Imblearn and Pandas to enhance data analysis. By integrating Matplotlib with Mlxtend, the process of visualizing data becomes more efficient. In addition, the Sklearn framework is implemented for data analysis. We have used the TensorFlow and Keras libraries in our research as a neural network library.

5 Result Discussion on CAN bus Datasets

This section presents specific procedures and methods implemented to demonstrate the proposed framework. This research compares the proposed IDS model to state-of-the-art approaches and examines it using a variety of performance measures to determine the effectiveness.

5.1 Evaluation Measures and Model Parameter Setup

This research used a metric combination of recall, precision, F1-score and accuracy to assess the finding of CPSO with B_LSTM network intrusion-detection model. These four metrics offer a thorough assessment of the proposed model's ability to accurately detect network intrusions in the IoV while minimizing the occurrence of both false positives and false negatives. Equations (14) to (19) represent the metrics of precision, recall, accuracy, F1 score, F2 score and Fbeta score.

5.1.1 Precision

The precision is characterized by the proportion of correctly predicted results relative to the overall count of correct predictions. A low false-positive rate is correlated with good accuracy. The precision is expressed as follows.

$$Precision(M_{pr}) = \frac{tp}{tp + fp} \quad (14)$$



5.1.2 Recall

A model's accuracy may be defined as the ratio of its true positive predictions to its total predictions inside a certain class. A strong recall is correlated with a low incidence of false negatives. The formula for recall is as follows.

$$\text{Recall}(M_{rc}) = \frac{tp}{tp + fn} \quad (15)$$

5.1.3 Accuracy

The most straightforward and precise metric for evaluating performance. The accuracy is the percentage of observations that were correctly predicted relative to the total number of observations. To determine precision, use the following formula:

$$\text{Accuracy}(M_{acc}) = \frac{tp + tn}{tp + tn + fp + fn} \quad (16)$$

5.1.4 F1 score

The harmonic mean of recall and precision is calculated. It is trying to attain equilibrium between precision and recall. An F1-score is considered optimal when it reaches a value of 1, whereas a model is regarded as unsuccessful when its F1-score is 0. The F1-score is computed using the following formula:

$$\text{F1score}(M_{f1}) = 2 * \frac{(M_{pr} * M_{rc})}{(M_{pr} + M_{rc})} \quad (17)$$

5.1.5 F2 score

The F2 score is calculated as the weighted harmonic mean of the precision and recall metrics. Contrary to the F1 score, which assigns equal importance to precision and memory, the F2 score attributes more importance to recall than to precision. The F2 Score is calculated as follows.

$$\text{F2score}(M_{f2}) = \frac{5 * (M_{pr} * M_{rc})}{4 * (M_{pr} + M_{rc})} \quad (18)$$

5.1.6 Fbeta score

In the computation of the harmonic mean, the Fbeta-measure is an abstraction of the F-measure in which the balance between precision and recall is regulated by a parameter

known as beta. The Fbeta score can be calculated as follows.

$$\text{Fbetascore}(M_{fbeta}) = \frac{1.25 * (M_{pr} * M_{rc})}{0.25 * (M_{pr} + M_{rc})} \quad (19)$$

Together, these performance criteria introduce a thorough evaluation of the model's performance. The precision and recall measures test the model's capacity to reduce both false positives and false negatives, while the accuracy measurement assesses the general functionality of the algorithm. The F1-score achieves a favorable equilibrium between recall and accuracy. IoV intrusion detection is dependent on these metrics. The model's precision in recognizing actual attacks allows us to minimize the occurrence of false alarms (false positives). To minimize the chances of detecting potential threats (false negatives) a high recall ensures that the system detects most intrusions. The F1-Score is a comprehensive assessment that considers all of them. The parameters employed for training the proposed B_LSTM + CPSO model, as well as experimental models including LSTM, LSTM + PSO, LSTM + FA, LSTM + CPSO, B_LSTM, B_LSTM + PSO and B_LSTM + FA, on all three CAN bus datasets, are shown in Table 7.

5.2 Result Discussion on Preliminary-Driving Dataset

Table 8 presents a comparison of the performance of each approach in detecting intrusions in IoV. The simulation of preliminary-driving Dataset demonstrates that the proposed CPSO with B_LSTM approach achieves optimum precision, recall, accuracy, F1 score, F2 score and Fbeta score compared to other techniques. The CPSO with B_LSTM approach demonstrated 94.42% accuracy, surpassing the standard B_LSTM method by 3.59%. The F1 score of the proposed CPSO with B_LSTM technique achieved a value of 0.9446, which was 0.0361 higher than the F1 score of the standard B_LSTM method. Compared to the proposed model, other approaches like LSTM + CPSO, B_LSTM + PSO and B_LSTM + FA demonstrate a satisfactory accuracy of 94.22%, 94.22% and 94.30%, respectively, which is the second highest performance. Figures 8 and 9 illustrate the evaluations of precision, recall, accuracy, F2 score, Fbeta score and F1 score for the Proposed B_LSTM with CPSO model. These evaluations are compared to other traditional models including the base methodology LSTM, B_LSTM and models that incorporate optimization approaches such as PSO, FA and CPSO, with LSTM and B_LSTM. Figure 10a, b shows the accuracy and loss plot of proposed CPSO with B_LSTM model. The analysis shows that the proposed model is outperforming the other approaches. Table 9 illustrates the analysis of a 10-fold accuracy investigation performed on the

Table 7 Considered parameters for model analysis

Methodology	Parameters
LSTM	Number of LSTM Cell Unit = 256, Activation Function = 'relu', Dropout Rate = 0.3, Learning Rate = 0.1, Optimizer = 'SGD'
LSTM + PSO	Number of LSTM Cell Unit = 256, Activation Function = 'tanh', Dropout Rate = 0.2, Learning Rate = 0.01, Optimizer = 'Adagrad'
LSTM + FA	Number of LSTM Cell Unit = 128, Activation Function = 'relu', Dropout Rate = 0.2, Learning Rate = 0.001, Optimizer = 'Nadam'
LSTM + CPSO	Number of LSTM Cell Unit = 128, Activation Function = 'relu', Dropout Rate = 0.15, Learning Rate = 0.01, Optimizer = 'Adam'
B_LSTM	Number of B_LSTM Cell Unit = 256, Activation Function = 'relu', Dropout Rate = 0.3, Learning Rate = 0.1, Optimizer = 'SGD'
B_LSTM + PSO	Number of B_LSTM Cell Unit = 64, Activation Function = 'relu', Dropout Rate = 0.3, Learning Rate = 0.01, Optimizer = 'Nadam'
B_LSTM + FA	Number of B_LSTM Cell Unit = 128, Activation Function = 'relu', Dropout Rate = 0.3, Learning Rate = 0.001, Optimizer = 'Adam'
Proposed B_LSTM + CPSO	Number of B_LSTM Cell Unit = 128, Activation Function = 'relu', Dropout Rate = 0.1, Learning Rate = 0.001, Optimizer = 'Adam'
Note : Fixed setup for all experimental models :	
Number of Layers = 3, Batch Size = 128,	
Loss Function = 'Categorical_crossentropy', Epoch = 160	

Table 8 Analysis of the proposed model in comparison with other state-of-the-art models on the preliminary-driving dataset

Methodology	$M_{acc}(\%)$	M_{pr}	M_{rc}	M_{f1}	M_{f2}	M_{fbeta}	GPU time elapsed (in Sec)
LSTM	89.62	0.8876	0.8834	0.8855	0.8854	0.8852	339.12
LSTM + PSO	93.58	0.9377	0.9392	0.9384	0.9382	0.9381	292.35
LSTM + FA	93.82	0.9385	0.9384	0.9384	0.9383	0.9383	279.43
LSTM + CPSO	94.22	0.9399	0.9424	0.9411	0.9410	0.9408	262.54
B_LSTM	90.83	0.9083	0.9087	0.9085	0.9083	0.9083	360.13
B_LSTM + PSO	94.22	0.9398	0.9425	0.9411	0.9410	0.9410	272.76
B_LSTM + FA	94.30	0.9432	0.9431	0.9431	0.9431	0.9429	242.51
Proposed B_LSTM + CPSO	94.42	0.9449	0.9444	0.9446	0.9445	0.9445	234.16

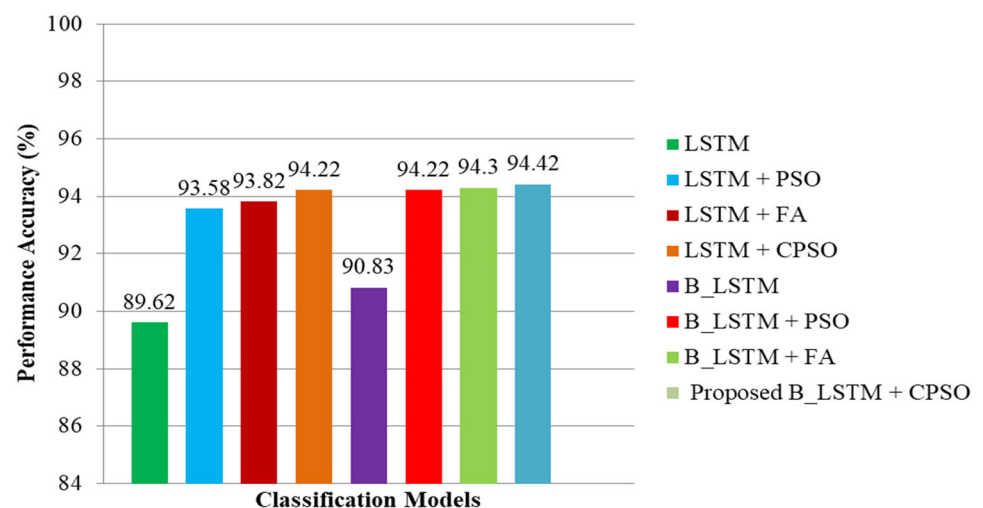
Fig. 8 Accuracy comparison of various DL approaches with proposed model on preliminary-driving dataset

Fig. 9 Performance comparison of different DL methodologies with proposed framework on preliminary-driving dataset

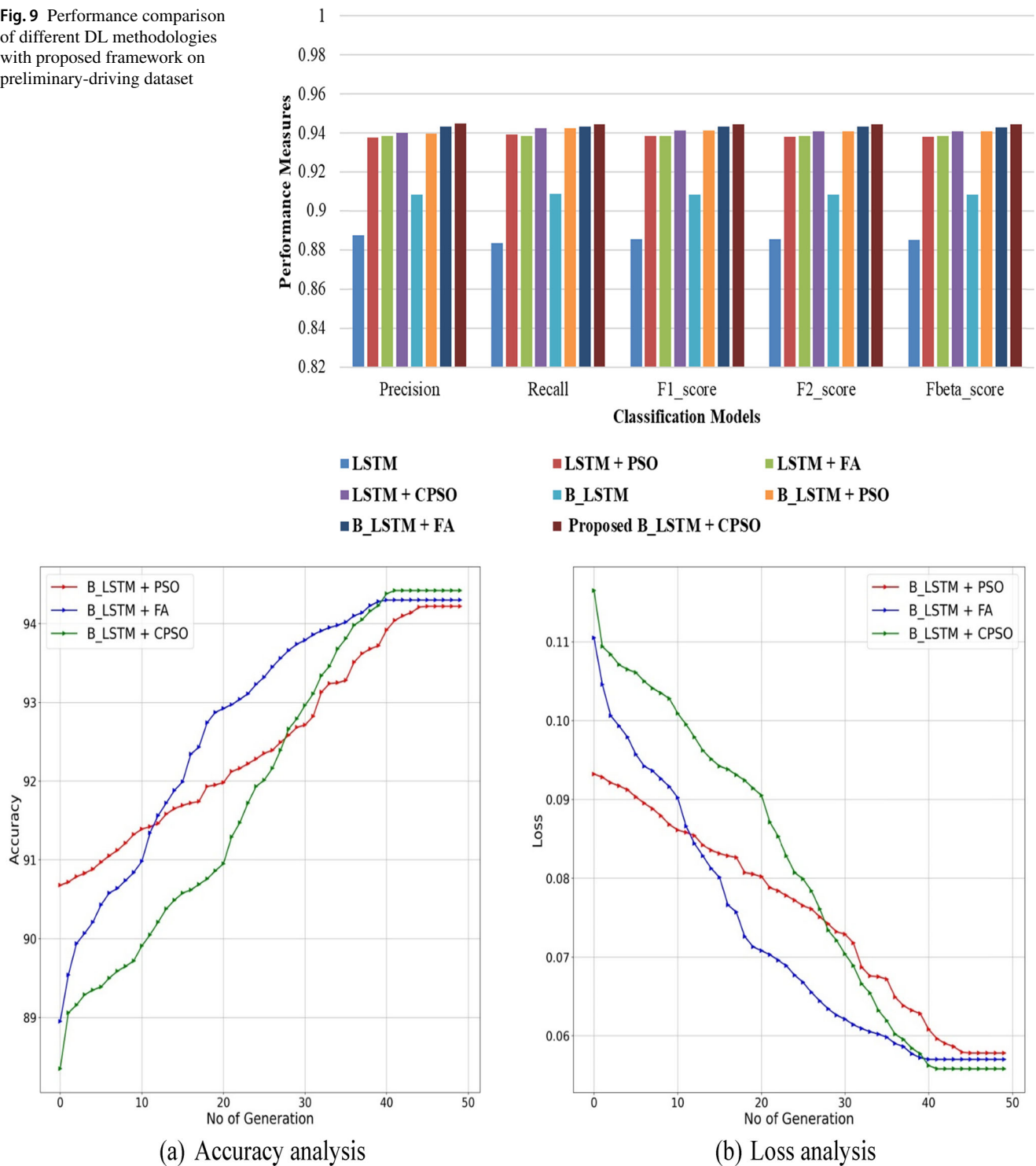
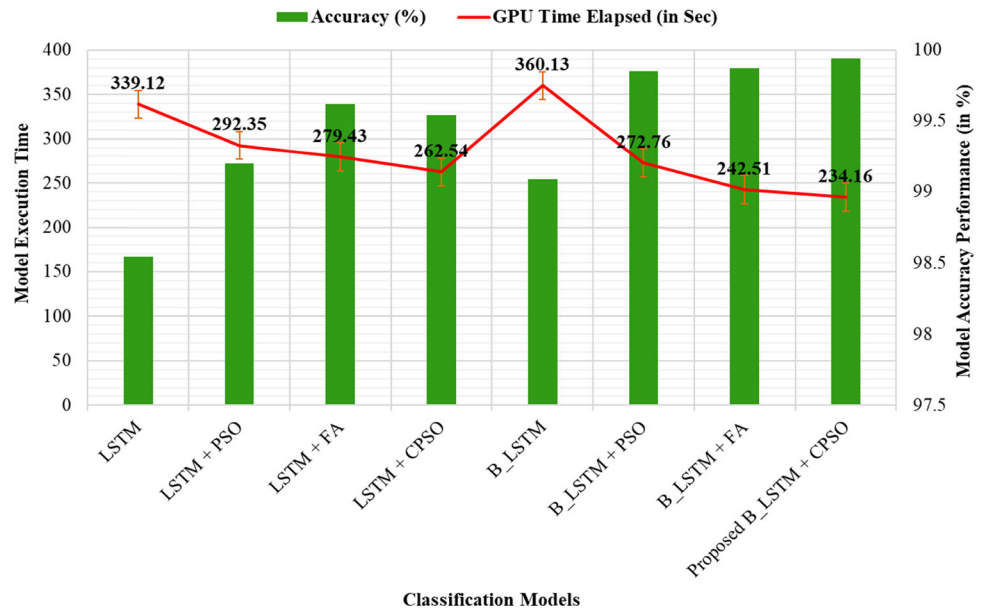


Fig. 10 Accuracy and loss analysis of on preliminary-driving dataset

Table 9 Tenfold accuracy validations of considered models on the preliminary-driving dataset

Tenfold	Tenfold accuracy measure (in %)							
	LSTM	LSTM + PSO	LSTM + FA	LSTM + CPSO	B_LSTM	B_LSTM + PSO	B_LSTM + FA	Proposed B_LSTM + CPSO
F1	89.62	93.58	93.82	94.22	90.83	94.22	94.30	94.42
F2	89.71	93.56	93.85	94.18	90.87	94.20	94.32	94.46
F3	89.66	93.51	93.82	94.26	90.85	94.21	94.30	94.42
F4	89.68	93.56	93.88	94.21	90.83	94.22	94.29	94.47
F5	89.60	93.58	93.89	94.22	90.86	94.17	94.30	94.44
F6	89.66	93.55	93.82	94.26	90.85	94.20	94.29	94.42
F7	89.61	93.59	93.85	94.22	90.83	94.21	94.30	94.46
F8	89.70	93.55	93.88	94.22	90.86	94.22	94.32	94.44
F9	89.62	93.58	93.82	94.20	90.85	94.20	94.33	94.42
F10	89.66	93.56	93.85	94.21	90.87	94.21	94.29	94.44
Average of tenfold	89.65	93.56	93.84	94.22	90.85	94.20	94.30	94.43

Fig. 11 Execution time analysis of experimental DL models on preliminary-driving dataset

Preliminary-Driving Dataset. The time elapsed for all experimental models on the Preliminary-Driving Dataset is shown in Fig. 11. It can be observed that the proposed B_LSTM + CPSO model has achieved optimum execution time on the GPU unit, which is 234.16 seconds.

5.3 Result Analysis on Preliminary Stationary Dataset

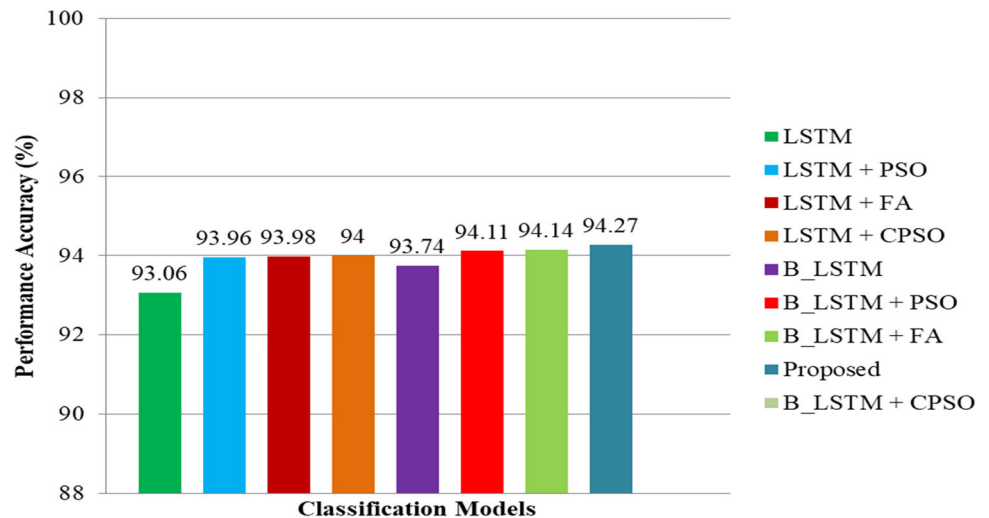
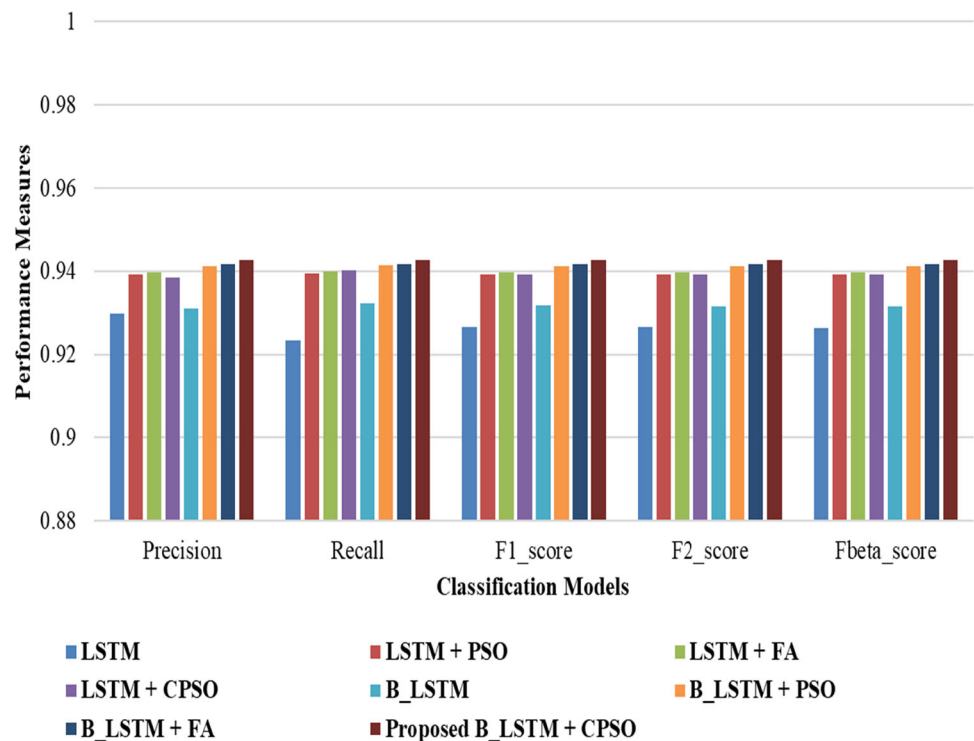
Table 10 shows a comparative analysis of the efficacy of each method in identifying intrusions in the IoV. The preliminary-stationary dataset simulation shows that the proposed CPSO

with B_LSTM method obtains the highest levels of precision, recall, accuracy, F1 score, F2 score and Fbeta score compared to other strategies. The CPSO with B_LSTM methodology obtained an accuracy of 94.27%, which was 0.53% higher than the normal B_LSTM method. The F1 score of the recommended CPSO with B_LSTM methodology reached a value of 0.9427, surpassing the F1 score of the normal B_LSTM method by 0.0110. When compared to the recommended model, additional techniques such as LSTM + CPSO, B_LSTM + PSO and B_LSTM + FA achieve an accuracy of 94.00, 94.11 and 94.14% correspondingly, which is the second highest level of performance. Figures 12 and



Table 10 Comparison presentation of proposed approach with other conventional models on preliminary-stationary dataset

Methodology	$M_{acc}(\%)$	M_{pr}	M_{rc}	M_{f1}	M_{f2}	M_{fbeta}	GPU time elapsed (in Sec)
LSTM	93.06	0.9298	0.9234	0.9266	0.9265	0.9263	336.71
LSTM + PSO	93.96	0.9392	0.9395	0.9393	0.9391	0.9392	297.32
LSTM + FA	93.98	0.9397	0.9399	0.9398	0.9397	0.9397	288.43
LSTM + CPSO	94.00	0.9385	0.9401	0.9393	0.9392	0.9391	276.09
B_LSTM	93.74	0.9311	0.9323	0.9317	0.9315	0.9316	381.42
B_LSTM + PSO	94.11	0.9412	0.9415	0.9413	0.9413	0.9412	294.53
B_LSTM + FA	94.14	0.9418	0.9417	0.9417	0.9416	0.9416	287.87
Proposed B_LSTM + CPSO	94.27	0.9427	0.9428	0.9427	0.9426	0.9427	256.33

Fig. 12 Accuracy analysis of various DL approaches with proposed model for preliminary-stationary dataset**Fig. 13** Performance measures analysis of various DL models with proposed model for preliminary-stationary dataset

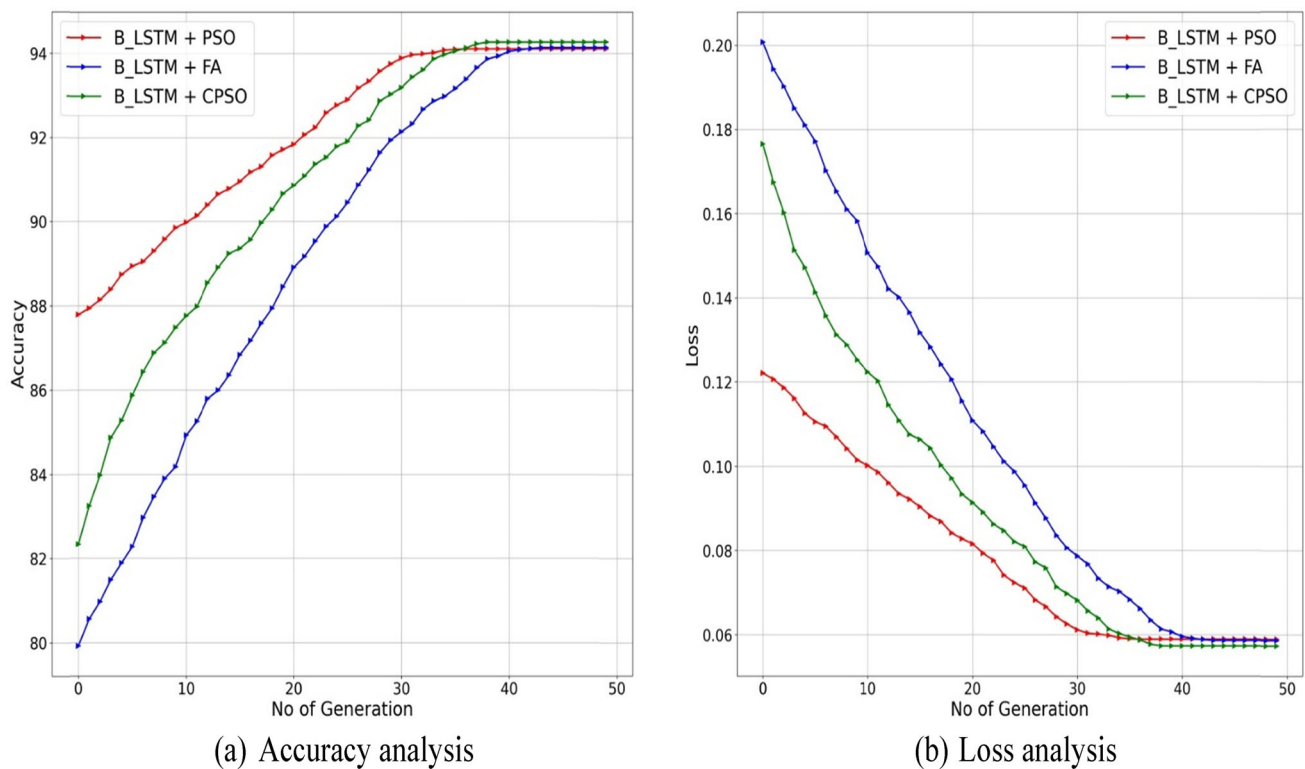


Fig. 14 Accuracy and loss analysis of on preliminary-stationary dataset

13 show the performance assessments of precision, recall, accuracy, F2 score, Fbeta score and F1 score for the Proposed B_LSTM with CPSO model. These assessments are compared to various advanced models, including the basic model LSTM, B_LSTM and models that include optimization methodologies such as PSO, FA and CPSO, along with LSTM and B_LSTM. Figure 14a, b shows the accuracy and loss plot of the proposed CPSO with B_LSTM model. This research indicates that the proposed model has performed better than the other models. Table 11 displays the results of a 10-fold accuracy study conducted on the preliminary-stationary dataset. The time elapsed for all experimental models on the preliminary-stationary dataset is presented in Fig. 15. We can again observe that the proposed B_LSTM + CPSO model has obtained optimal execution time on the GPU unit of 256.33 seconds.

5.4 Discussion on Final Stationary Dataset Findings

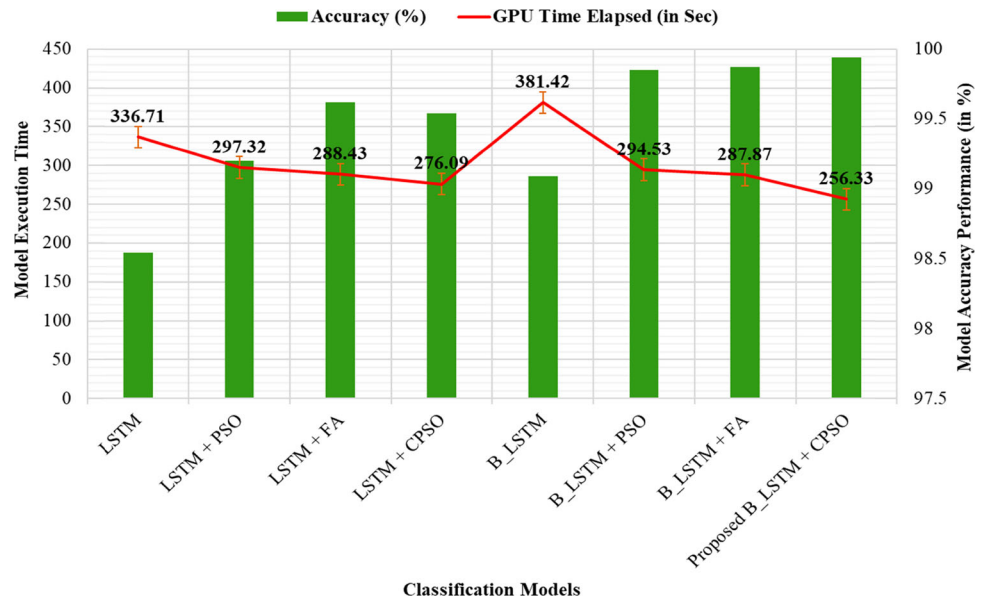
Table 12 presents a compared assessment of the effectiveness of each strategy in detecting intrusions in the IoV. The final-stationary dataset simulation demonstrates that the CPSO with B_LSTM approach achieves superior levels of precision, recall, accuracy, F1 score, F2 score and Fbeta score in comparison with other methods. The CPSO using the

B_LSTM approach obtained an accuracy of 99.94%, surpassing the regular B_LSTM method by 0.85%. The F1 score achieved by the proposed CPSO with B_LSTM approach was 0.9988, overcoming the F1 score of the standard B_LSTM method by 0.093. When compared to the proposed model, the additional approaches of LSTM + FA, B_LSTM + PSO and B_LSTM + FA obtain accuracies of 99.62%, 99.85% and 99.87%, respectively. These accuracies represent the second highest level of performance. Figures 16 and 17 show the experimental evaluations of precision, recall, accuracy, F2 score, Fbeta score and F1 score for the Proposed B_LSTM with CPSO model. These evaluations are compared to many sophisticated models, including the fundamental model LSTM, B_LSTM and models that include optimization approaches such as PSO, FA and CPSO, in addition to LSTM and B_LSTM. Figure 18a, b shows the accuracy and loss plot of the proposed CPSO with B_LSTM model. The investigation suggests that the proposed model has outperformed the other models. Table 13 shows the results of a 10-fold accuracy test on the final-stationary dataset. Figure 19 illustrates the total amount of time that elapsed for all experimental models on the final-stationary dataset. The optimal execution time of the proposed B_LSTM + CPSO model on the GPU unit is thus 227.39 seconds.



Table 11 Tenfold accuracy validations of considered models on the preliminary-stationary dataset

Tenfold	Tenfold accuracy measure (in %)							
	LSTM	LSTM + PSO	LSTM + FA	LSTM + CPSO	B_LSTM	B_LSTM + PSO	B_LSTM + FA	Proposed B_LSTM + CPSO
F1	93.10	93.94	93.99	94.00	93.72	94.11	94.14	94.27
F2	93.05	93.92	93.98	94.10	93.73	94.13	94.10	94.21
F3	93.06	93.94	93.94	94.06	93.77	94.15	94.11	94.22
F4	93.09	93.98	93.96	94.04	93.74	94.09	94.15	94.24
F5	93.10	93.96	93.94	94.00	93.72	94.03	94.17	94.21
F6	93.08	93.94	94.03	94.03	93.73	94.11	94.14	94.23
F7	93.13	93.95	93.98	94.00	93.77	94.18	94.18	94.28
F8	93.04	93.96	93.98	93.98	93.79	94.16	94.20	94.25
F9	93.06	93.95	93.96	94.10	93.78	94.17	94.14	94.22
F10	93.08	93.96	93.93	94.03	93.74	94.13	94.11	94.27
Average of 10-Fold	93.07	93.95	93.96	94.03	93.74	94.12	94.14	94.24

Fig. 15 Execution time analysis of experimental DL models on preliminary-stationary dataset**Table 12** Comparison with proposed model vs state-of-the-art models on the final-stationary dataset

Methodology	$M_{acc}(\%)$	M_{pr}	M_{rc}	M_{f1}	M_{f2}	M_{fbeta}	GPU time elapsed (in Sec)
LSTM	98.54	0.9751	0.9787	0.9769	0.9768	0.9766	369.46
LSTM + PSO	99.20	0.9895	0.9905	0.9900	0.9898	0.9894	336.75
LSTM + FA	99.62	0.9965	0.9963	0.9964	0.9963	0.9962	279.60
LSTM + CPSO	99.54	0.9865	0.9895	0.9880	0.9878	0.9877	249.31
B_LSTM	99.09	0.9901	0.9890	0.9895	0.9894	0.9894	371.65
B_LSTM + PSO	99.85	0.9915	0.9945	0.9930	0.9928	0.9929	304.55
B_LSTM + FA	99.87	0.9988	0.9987	0.9987	0.9986	0.9986	292.61
Proposed B_LSTM + CPSO	99.94	0.9989	0.9988	0.9988	0.9987	0.9988	227.39



Fig. 16 Accuracy analysis of various DL methodologies with proposed model on final-stationary dataset

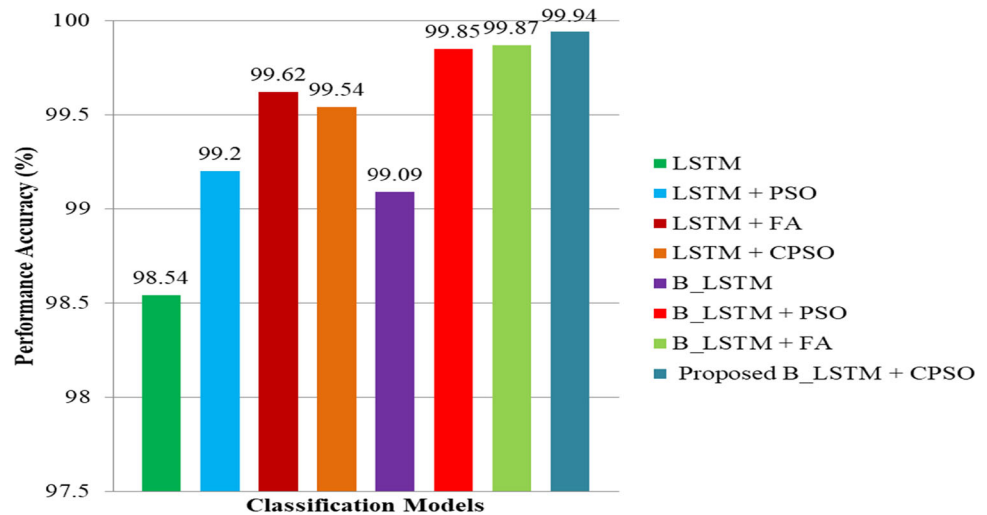
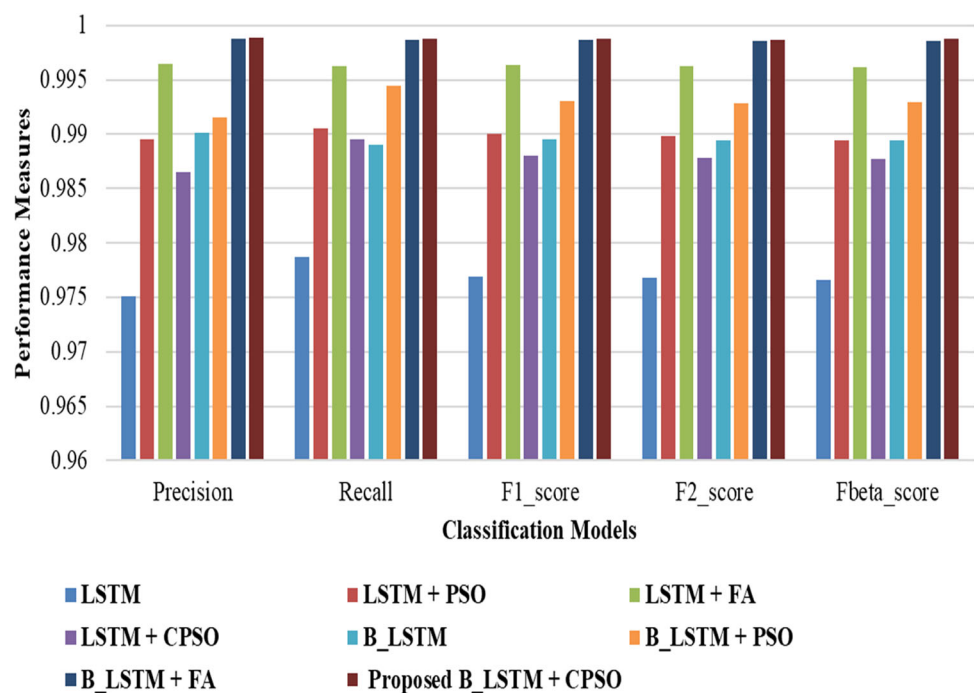


Fig. 17 Performance measures analysis of experimented DL models with proposed model on final-stationary dataset



5.5 Limitation and Discussion

The IoV environment is defined by network traffic data that exhibits large-scale, diverse numerous sources, high-dimensionality and temporal correlation. Conventional IDS need several nonlinear transformations and complex matrix manipulations for processing network traffic data, therefore minimizing their accuracy and performance. The present research employs the B_LSTM + CPSO architecture to develop an IDS that minimizes nonlinear transformations and matrix manipulations. The B_LSTM + CPSO framework has the capability to effectively identify relationships within temporal correlation data. Using the benefits of the B_LSTM + CPSO architecture, the proposed method can

improve the overall performance in handling large-scale, diverse multiple sources, high-dimensionality and temporal correlation IoV network traffic data. IoV datasets often show severe class imbalance with intrusion activities occurring very rarely compared to normal network traffic. This mismatch may cause model bias which increasing the risk of false negatives when attacks are not detected. Addressing this imbalance is crucial for increasing model performance.

Furthermore, noise in IoV data may be caused by a variety of sources, including sensor failures, communication interruptions and environmental interference. Effectively regulating this noise is a significant challenge for the B_LSTM+CPSO framework. Time needed for the proposed model to analyze incoming IoV traffic data and identify



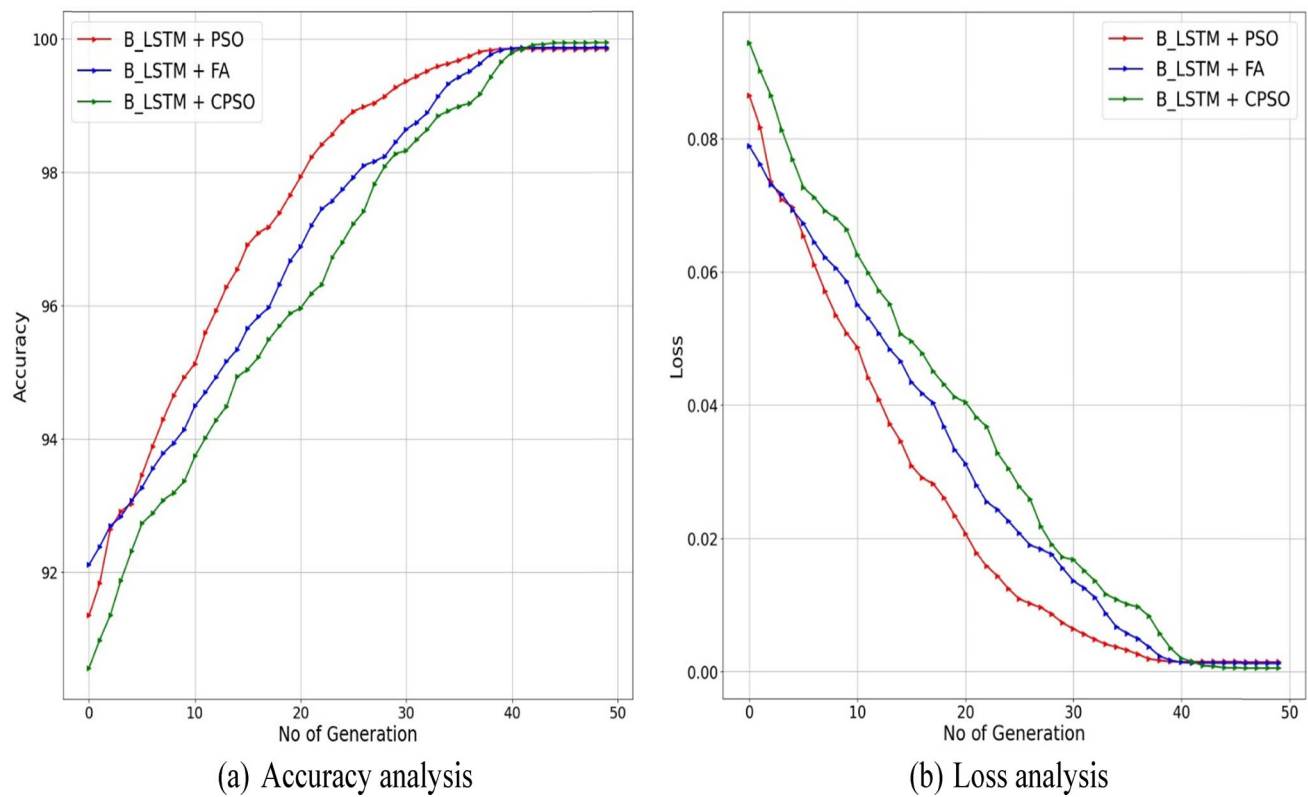
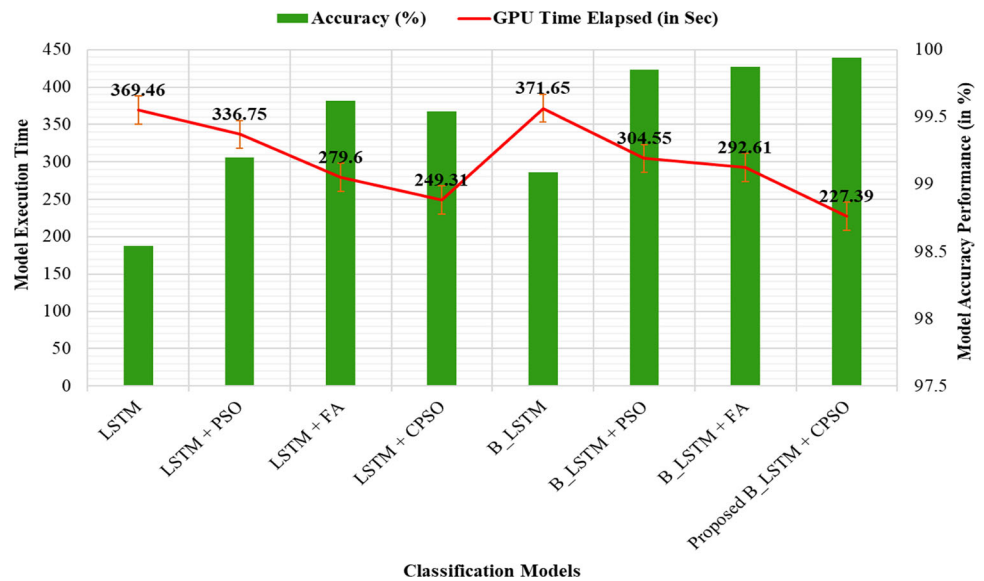


Fig. 18 Accuracy and loss analysis of on final-stationary dataset

Table 13 Tenfold accuracy validations of considered models on the final-stationary dataset

Tenfold	Tenfold accuracy measure (in %)				B_LSTM	B_LSTM + PSO	B_LSTM + FA	Proposed B_LSTM + CPSO
	LSTM	LSTM + PSO	LSTM + FA	LSTM + CPSO				
F1	98.54	99.20	99.62	99.54	99.09	99.85	99.87	99.94
F2	98.56	99.18	99.60	99.53	99.04	99.84	99.88	99.93
F3	98.62	99.16	99.66	99.52	99.08	99.85	99.87	99.98
F4	98.56	99.23	99.62	99.50	99.09	99.85	99.84	99.91
F5	98.51	99.20	99.64	99.56	99.05	99.86	99.87	99.93
F6	98.54	99.18	99.66	99.54	99.06	99.84	99.86	99.94
F7	98.56	99.19	99.62	99.53	99.07	99.83	99.85	99.92
F8	98.60	99.20	99.68	99.52	99.08	99.86	99.87	99.93
F9	98.54	99.17	99.63	99.55	99.09	99.85	99.86	99.92
F10	98.56	99.20	99.61	99.54	99.04	99.84	99.87	99.94
Average of tenfold	98.55	99.19	99.63	99.53	99.06	99.84	99.86	99.93

Fig. 19 Execution time analysis of experimental DL models on final-stationary dataset



intrusions may result in delay, which is crucial in IoV environment where prompt decisions are essential. Under dynamic conditions, this proposed model may need regular improvements to maintain its effectiveness. The retraining of the B_LSTM+CPSO model may lack sufficient speed for real-time IoV security problems. The successful implementation of this proposed model should be evaluated not just in simulated environments or with a specific dataset, but also in real-life applications of the IoV. This may include collaborating with automobile manufacturers or independent vehicle service providers to conduct real-world experiments of the model. Moreover, Exploring the incorporation of adaptive techniques that enable the model to autonomously update itself without the need for explicit retraining. This may include the use of online learning methodologies or model pruning to decrease the computing complexity.

5.6 Comparison with Existing Studies

Over the years, several methods have been proposed to create effective IDS for the IoV network. The techniques adopted vary from conventional approaches to more complex computational models such as neural networks (NNs). The detection of attacks inside the IoV network can potentially be achieved by using frameworks such as CNN, LSTM and B_LSTM. The literature review section offers an extensive overview of the results from several prior studies on cyber-attacks in IoV environments. Table 14 summarizes the findings of a prior research obtained from several CAN bus datasets that are chosen for evaluation. Prior studies have shown that several classification measures have achieved high levels of accuracy. However, the proposed methodology produces the results within an improved range. Therefore, it has been obtained that the proposed CPSO with B_LSTM approach

shown better performance compared to the prior research and approaches that were investigated.

5.7 Statistical Analysis

A widely recognized statistical procedure has been performed to quantitatively analyze the performance of all the models that have been evaluated in this research. The performance evaluation findings are shown in Table 15. While performing the execution of these experiments, null hypothesis have established which states that all modifications and resemblances in performance across the learning models are purely caused by coincidence and hence they are the same. The average rankings of models are determined using the Friedman Test shown in Table 15 and calculated using equation (20). In this scenario, the lowest rank (i.e., 1) is given to the highest fitness value. All variables represented by are wrapped in curly brackets and the corresponding values of are shown in the last row of Table 15.

$$Rank_j = \frac{1}{N} \sum_i rank_i^j \quad (20)$$

$$Z_F^2 = \frac{12N}{m(m+1)} \left[\sum_i Rank_j^2 - \frac{m(m+1)^2}{4} \right] \quad (21)$$

$$Frieds = \frac{(N-1)Z_F^2}{N(m-1) - Z_F^2} \quad (22)$$

$$Z = \frac{(Rank_i - Rank_j)}{\sqrt{m(m+1)/6N}} \quad (23)$$

Among 'N' performance measures, 'Rank_j' indicates the average rank of the jth model, while 'rank_i^j', indicates the



Table 14 Comparison of proposed model with existing studies

Year	Dataset used	Model used	Accuracy (%)	Ref
2020	IVN	LSTM Autoencoder	99.00	[18]
	UNSW-NB15		98.00	
2021	OTIDS	RF	98.10	[14]
2021	UNSW-NB15	B_LSTM	99.88	[20]
	Automobile Hacking		99.11	
	CICIDS 2017		99.64	
2022	UNSW-NB15	CNN-LSTM	94.53	[22]
	WSNDS		99.67	
	CAN bus final stationary		99.94	
2024	CAN bus final stationary	B_LSTM + CPSO	99.94	Proposed work

Table 15 Friedman's rank of DL classifiers based on various performance metrics

Stratified sampled IoV data	Tenfold accuracy measure (in %)							Proposed B_LSTM + CPSO
	LSTM	LSTM + PSO	LSTM + FA	LSTM + CPSO	B_LSTM	B_LSTM + PSO	B_LSTM + FA	
SF1	98.54(8)	99.20(6)	99.62(4)	99.54(5)	99.09(7)	99.85(3)	99.87(2)	99.94(1)
SF2	98.56(8)	99.18(6)	99.60(4)	99.53(5)	99.04(7)	99.84(3)	99.88(2)	99.93(1)
SF3	98.62(8)	99.16(6)	99.66(4)	99.52(5)	99.08(7)	99.85(3)	99.87(2)	99.98(1)
SF4	98.56(8)	99.23(6)	99.62(4)	99.50(5)	99.09(7)	99.85(2)	99.84(3)	99.91(1)
SF5	98.51(8)	99.20(6)	99.64(4)	99.56(5)	99.05(7)	99.86(3)	99.87(2)	99.93(1)
SF6	98.54(8)	99.18(6)	99.66(4)	99.54(5)	99.06(7)	99.84(3)	99.86(2)	99.94(1)
SF7	98.56(8)	99.19(6)	99.62(4)	99.53(5)	99.07(7)	99.83(3)	99.85(2)	99.92(1)
SF8	98.60(8)	99.20(6)	99.68(4)	99.52(5)	99.08(7)	99.86(3)	99.87(2)	99.93(1)
SF9	98.54(8)	99.17(6)	99.63(4)	99.55(5)	99.09(7)	99.85(3)	99.86(2)	99.92(1)
SF10	98.56(8)	99.20(6)	99.61(4)	99.54(5)	99.04(7)	99.84(3)	99.87(2)	99.94(1)
Avg. Friedman's Rank	8	6	4	5	7	2.9	2.1	1

exact rank of the j th model among a collection of ' m ' models for the i th measure. Equation (21) produces a value of 69.70 for the ' Z_F^2 ' statistic after the calculation of the average rank. Then, applying Eq. (22), the Friedman statistics ' $Fried'_S$ ' is being calculated as 209.10. The critical value of 2.70 was found and a significance level of $\alpha = 0.01$ (see Fig. 20) based on the Friedman statistics $Fried'_S = 209.10$ with degrees of freedom (9, 63) in this investigation. The research finds that the ' $Fried'_S$ ' statistics observed value is less than the key value of 2.70, therefore rejecting the null hypothesis (N_{H_0}).

The Holm approach utilizes ' Z ' values, which are computed from the mean rankings of models as shown in Table 15 using Eq. (23). Subsequently, the ' p ' values are computed by using the normal distribution table using the obtained ' Z ' values. Table 16 describes a thorough comparison of the analyzed models, emphasizing their calculated ' Z ' values, ' p ' values and the value $\alpha / (k - 1)$. In this particular context, the variable ' i ' denotes the specific model number,

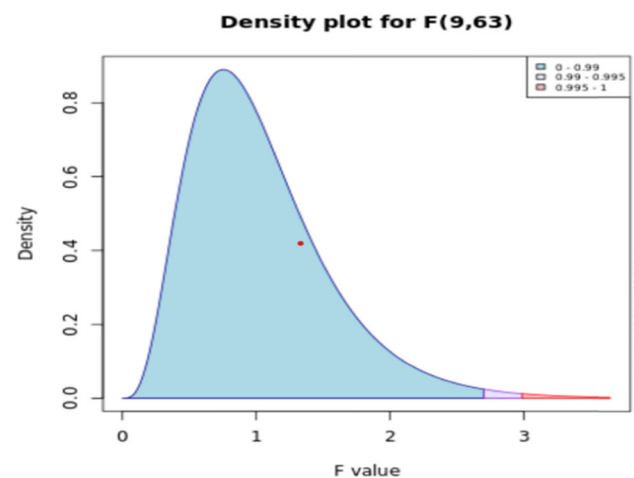
**Fig. 20** Statistical analysis density plot

Table 16 The results of the Holm and Hochberg test

<i>I</i>	1	2	3	4	5	6	7
Learning Replica	B_LSTM + CPSO: LSTM	B_LSTM + CPSO: LSTM + PSO	B_LSTM + CPSO: LSTM + FA	B_LSTM + CPSO: LSTM + CPSO	B_LSTM + CPSO: B_LSTM	B_LSTM + CPSO: B_LSTM + PSO	B_LSTM + CPSO: B_LSTM + FA
<i>z</i> value	6.363636	4.545454	2.727272	3.636363	5.454545	1.727272	1
<i>p</i> value	0.0000000001	0.0000027435	0.0031930714	0.0001382919	0.0000000246	0.0420593747	0.1586552594
$\frac{\alpha}{k-1}$	0.00142	0.00166	0.002	0.0025	0.00333	0.005	0.01

while the symbol ' α ' is equivalent to the numerical value of 0.01. Consequent, the null hypothesis (N_{H_0}) being examined is rejected. It is well noticed that in most circumstances, the value of ' p_i ' is lower than $\alpha / (k - 1)$. The variable ' Z ' represents the ' $Z - score$ ' rate, ' k ' indicates the number of models, ' N ' represents the number of performance measures and ' $Rank_i$ ' and ' $Rank_j$ ' reflect the average rank of the ' i th' and ' j th' models, respectively.

6 Conclusions and Future Scope

The primary goal of this research was to enhance the efficiency of B_LSTM in IoV intrusion detection by using effective hyper-parameter tuning. This research aimed to optimize the hyper-parameters of the B_LSTM model by using three well recognized evolutionary computing algorithms: PSO, FA and CPSO. By conducting rigorous experiments on three significant CAN bus dataset such as the final stationary dataset, the preliminary stationary dataset and the preliminary driving dataset, this research illustrated that the proposed models have the potential to significantly enhance the effectiveness of B_LSTM in detecting intrusions in the IoV. The results confirmed that effective hyper-parameter adjustment may lead to significant improvements in model performance, strengthening its crucial position in DL models for IoV intrusion detection. By incorporating the techniques of FA, PSO and CPSO for hyper-parameter optimization, it has been observed that the B_LSTM model exhibits a significant improvement in detection accuracy and a decrease in false-positive rates. This improvement is evident when comparing the B_LSTM model to other basic LSTM models, such as LSTM+FA, LSTM+PSO, LSTM+CPSO, B_LSTM+FA and B_LSTM+PSO models, all with selected hyper-parameter configurations. However, the degree of enhancement varied across the datasets and was impacted by the characteristics of the data and the specific type of attack available in each dataset. Therefore, there was no universally better approach between FA, PSO and CPSO, since the most successful algorithm differed depending on the dataset. Hence, the selection of the optimization

technique must be adapted to the individual circumstances, highlighting the need of having an in-depth understanding of the intended datasets.

6.1 Future Scope

In future, further research might examine a wider range of optimization approaches or explore into the potential of multi-objective optimization to enhance the model's performance. Furthermore, given the significant privacy issues in IoV settings, it would be beneficial for future study to explore the integration of privacy-preserving methods, such as federated learning (FL) into the CPSO-optimized BLSTM framework. This integration would ensure safe data processing over distributed vehicle networks. In addition, the CPSO-optimized BLSTM model may be combined with other DL models such as reinforcement learning or generative adversarial networks (GANs) to develop more integrated and advanced IDS for the IoV.

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Declarations

Conflict of interest The authors declare that this manuscript has no conflict of interest with any other published source and has not been published previously (partly or in full). No funding was received for conducting this study.

Human Participants and Animals This study does not involve any human participants or animals for research.

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